LAKE CHILWA BASIN CLIMATE CHANGE ADAPTATION PROGRAM

Livelihoods in the Lake Chilwa Basin

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Finally, the findings presented here are not views of the author but those from the analysis of interviews conducted during the study.

ACRONYMS

GOM	Government of Malawi
IPCC	Intergovernmental Panel on Climate Change
NGO	Non-Governmental Organization
CBO	Community Based Organization
LCBCCAP	Lake Chilwa Basin Climate Change Adaptation Program
VNRMC	Village Natural Resources Management Committee
FAO	Food and Agricultural Organization of the United Nations
FGD	Focus Group Discussion
WUA	Water Users Association
ADC	Area Development Committee
VDC	Village Development Committee
GVH	Group Village Head
VH	Village Head
DFID	Department for International Development of the United Kingdom

SUMMARY

A livelihoods study was undertaken to understand the current household livelihoods options of local communities residing in the Lake Chilwa Basin and their capacity to adapt to climate variability in an effort to reduce vulnerability and poverty. Past livelihoods and ecological studies have shown that the Lake Chilwa Basin constitutes a fragile ecosystem based on the extreme pressure from deforestation and fires in its catchment, as well as the periodic complete desiccation of the lake on several occasions in the past (1913-16, 1922, 1934, 1943-48, 1967, 1973 and 1995). Although people adapted and learned to cope with these continuously changing and fragile environments in order to sustain their livelihoods, the adaptation strategies they adopted may have caused devastating effects on the environment and local economy.

The approach used for the livelihoods survey was based on the framework and principles of the DFID Sustainable Livelihoods Approach (SLA). As such the study relied on a combination of quantitative and qualitative methods data collection tools. Focus Group Discussions (FGD) were conducted in fourteen villages using *Resource mapping, Institutional analysis: matrix scoring and ranking; Venn diagrams, Cause-effect analysis and Seasonal calendars.* A semi-structured questionnaire was administered to about 900 respondents. Additional key informant interviews were also conducted with some members of the communities including secondary information.

The analysis of livelihoods of the Lake Chilwa Basin communities highlighted the following issues:

- Weak livelihoods capital assets particularly natural, financial, human and physical resources. Only social capital can be considered as strong in some cases.
- Weak policy framework that determines access to natural resources. The conceptualisation of community based approaches that has fostered ideas about community-based conservation, is not adequately implemented as such communities' participation in natural resource management is passive.
- High livelihoods vulnerabilities to guarantee sustainable natural resource management. There is low livelihoods diversity coupled with over-dependency on natural resources. Unfortunately many livelihood options are very seasonal. The period January to February is so critical where household incomes are very low. However cash demand for farm operations is at peak. Additionally the basin is prone to three sanitary-related diseases of malaria, diarrhea and bilharzia. Malaria prevalence stems from two factors: the presence of a swamp and the misuse of bed nets. Diarrhea is caused by lack of safe

and clean drinking water. For example, during severe cholera outbreaks 40–45 percent of inhabitants are affected.

There is however scope for improving livelihoods even in contexts where natural resource dependence is high through better choice of strategies, diversified and enduring partnerships, a better representation of communities leading to an active role in the preparation and implementation of policies. These could be achieved through building social capital for empowerment, strong financial services, and resilience of communities to climate variability and change. There is also need to add value to existing commodities in the basin namely: fish, rice, pigeon peas, cassava, sunflower and firewood. Formulation of strategies and actions as interventions for the identified policy gaps will lead to successful conservation, management and development of the natural resources in the Lake Chilwa Basin thereby enhancing communities' adaptive capacity to climate variability.

1. INTRODUCTION

The Lake Chilwa Basin is shared between Malawi and Mozambique with 68% (5,669 km^2) in Malawi and 32% (2,680 km^2) in Mozambique (GoM, 1999; GoM, 2001). The 2008 population census shows that the catchment area has approximately 1.6 million people with a density of 321 people per km^2 registering an increase of approximately 50% from that of 1998 (NSO, 2008). The population figures indicate that the Lake Chilwa catchment is one of the most densely populated areas of Malawi. Most of the basin's population (90 %) relies on fishing, farming and petty trading on fish and farm products for their livelihoods (Zimba & Kaunda, 1999). The Basin is biologically an important wetland classified as a Ramsar site in 1997. Literature shows that there have been efforts to declare the Lake Chilwa wetland a protected area. The British Central African Administration's efforts to declare the wetland a game reserve in 1897, 1902 and 1911 proved futile (GoM, 2001). Currently the basin provides diverse economic benefits to its inhabitants. The Lake Chilwa itself contributes up to $17 million annually from fish catches and fishing provides greater economic returns than farming per sq km (Schuijt, 1999). The lake and the whole of the Chilwa plains are an economic network. Not only are there links between fishing and various ancillary services, but also complementary flows of income between fishing, farming and cattle-rearing (Phipps 1973, Kalk *et al*, 1979).

Past livelihoods and ecological studies (Allison & Mvula 2002; Lowore & Lowore (1999); Landes & Otte (1983); Kalk *et al*, 1979; Njaya 2009; Sarch & Allison 2000) have shown that the Lake Chilwa Basin constitutes a fragile ecosystem based on the extreme pressure from deforestation and fires in its catchment, as well as the periodic complete desiccation of the lake on several occasions in the past. The lake dried up eight times: in 1903, 1913-16, 1922, 1934, 1943-48, 1967, 1973 and 1995 (GoM, 2000a). Although people depending on the basin's natural resources adapted and learned to cope with these continuously changing and fragile environments in order to sustain their livelihoods, these dry-ups may have caused devastating effects on the local economy.

It is clear that climate change will, in many parts of the world, adversely affect socio-economic sectors, which include water resources, agriculture, forestry, fisheries, ecological systems and human settlements and health. Developing countries are the most vulnerable (IPCC 2007). The Lake Chilwa Basin community is such an example that may be affected highly as its livelihoods is structured on natural resource use. Therefore, this livelihoods study aimed at understanding the current livelihoods strategies and the capacity of households to adapt to climate variability in an effort to reduce vulnerability and poverty of local communities residing in the basin.

A livelihood in this study comprises of the capabilities, assets (including both material and social resources), and activities required for a means of living. Communities with better livelihoods have strong asset base (e.g. good health and education) and secured access to natural resourceshence better adaptive capacities. The concept of access is central in understanding how distribution of power is structured, how it determines the distribution of the means of production, and ultimately how it influences people's livelihood opportunities as well as sustainability of natural resources. Sociopolitical processes are said to affect peoples' access and use of resources. In recognition of this, Sen (1996) argues that starvation is the characteristic of some people not having access to enough food, not a characteristic of there not being enough food to eat. Furthermore, a livelihood is sustainable when it can cope with and recover from stresses and shocks and maintain or enhance its capabilities and assets both now and in the future, while not undermining the natural resource base' (Chambers, and Conway, 1992). The Brundtland Commission in 1987 introduced sustainable livelihoods in terms of resource ownership and access to basic needs and livelihood security, especially in rural areas.

This report analyses the status of these assets and how they are accessed by the Lake Chilwa communities. Section 2 provides the tools used to assess these assets. Results of the survey are presented in section 3 and recommendations are provided in section 4.

2. METHODOLOGY

The approach used for the livelihoods survey was based on the framework and principles of the sustainable livelihoods approach (SLA). Conceptually, “livelihoods” connote the means, activities, entitlements and assets by which people make a living. Assets, in this particular context, are defined as not only natural/biological (i.e., land, water, common-property resources, flora, fauna), but also social (i.e., community, family, social networks, participation, empowerment, human (i.e., knowledge, creation by skills) and physical (i.e., roads, markets, clinics, schools, bridges). As such the study relied on a combination of quantitative and qualitative methods of data collection.

The participatory study was organised in the following main phases**:**

- Preparatory phase
- Exploratory research
- Qualitative data collection
- Quantitative data collection
- Data analysis

2.1 Preparatory phase

The preparatory phase started with the development of study approach that included steps for carrying out the survey (appendix 1). Tools for the study were then developed that included qualitative study guidelines (appendix 2) and semi-structured questionnaire (appendix 3). Two teams were then assembled to carry out the quantitative and qualitative data collection respectively.

The study targeted specific hotspots within three major agro-ecological zones namely: the catchment, the flood plain and the lake to take care of the possible variation in livelihoods sources. Through a participatory process with district administration staff, 21 villages covering the three zones above were selected as study sites (Figure 1).

2.2 Exploratory phase

This stage consisted of the review of the available secondary information that included reports and publications on the main sectors: fisheries, agriculture, forestry, water and government policies. The main objective of this work was to identify the strong points and gaps from previous studies in the basin from which data could be taken to improve the understanding of people’s livelihoods in the basin.

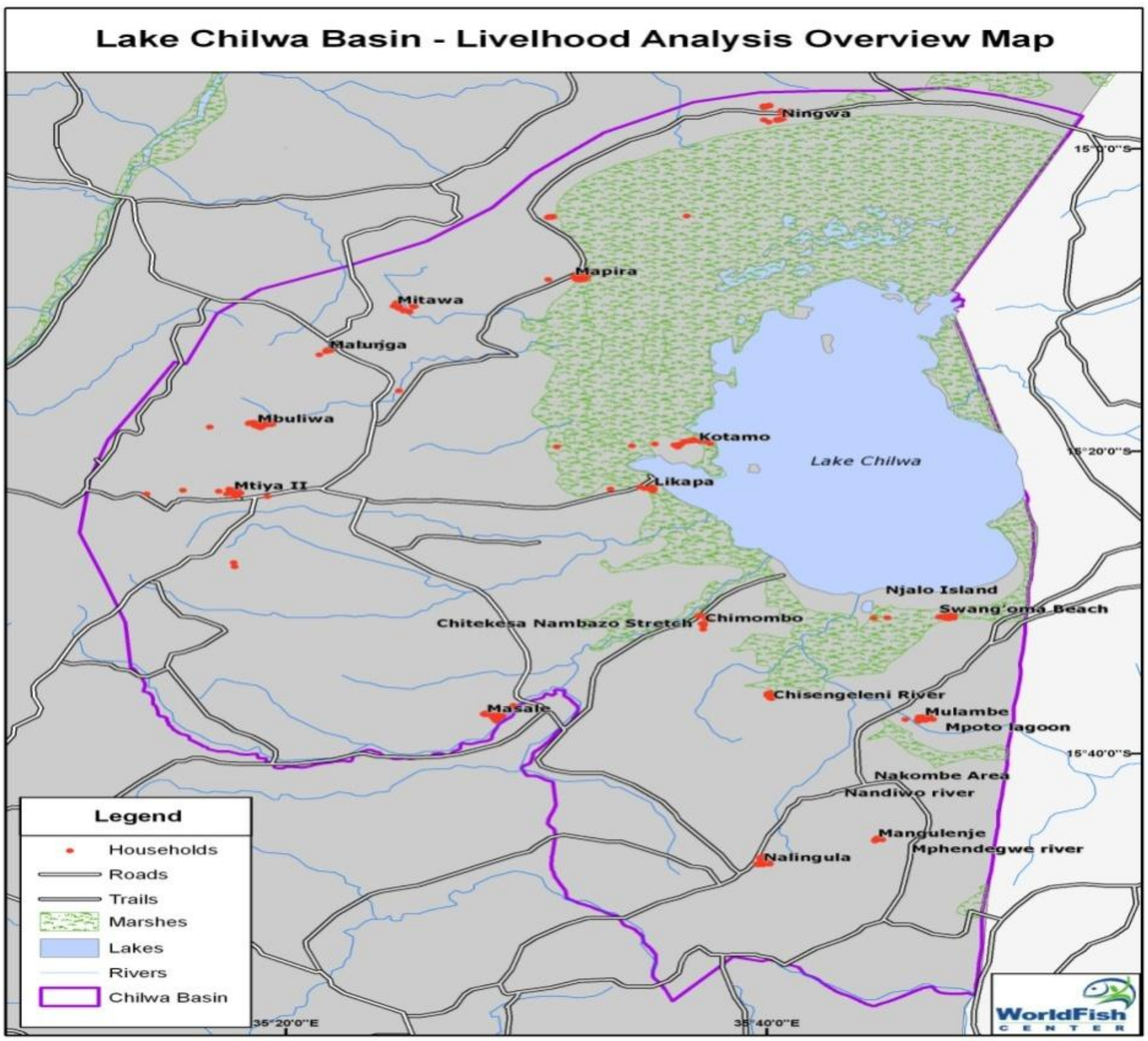


Figure 1: Livelihoods Overview Map – LCBCCAP

2.3 Qualitative study

This phase comprised of participatory diagnoses of the livelihoods of selected villages in the Lake Chilwa Basin, complemented by collection of socio-economic data. Four broader questions were asked through focus group discussions in 12 villages covering the upstream, middle stream and lower stream on the Lake Chilwa Basin. These questions were:

- How do people define well-being or good quality of life and ill-being or bad quality of life?
- How do people perceive vulnerability, risk, security and opportunities? and how they have changed over time?
- How do households cope with possible decline in well-being? How do these coping strategies in turn affect their lives?
- What are people's hopes and fears for the future?

To answer these questions the following tools were used:

Well-being trend analysis:

This tool was used to analyze livelihoods changes that have taken place in the community and people's perceptions about the future. The group was asked to define well-being. After the definition was agreed by both women, men and the youth, the whole community was categorized into three: well-off, average and poor. Changes in the numbers of people per category over the last 10 years and respective contributing factors were discussed. The groups were also asked on their perceptions of the future for the categories and the contributing factors.

Resource mapping

The group was asked to draw a big map of their village. As they were drawing discussions on other details of the map were done and in general included the following questions: Which places are important for the people of this community? Do different groups (men, women fishers, farmers, hunters etc) have different places that are important to their lives? Why are these places important? What natural resources are available in this community now? What natural resources have changed in the past 10 years? What sources of livelihood have changed due to changes in natural resources? Are there any customs, rules and norms in this community that facilitate or hinder access to natural resources? What needs to be done if livelihoods have to improve?

Institutional analysis: matrix scoring and ranking; Venn diagrams

This aimed at exploring the most important formal, informal, government, and non-government institutions within or outside the community that influence people's lives. Institutions play a critical role in supporting or constraining people's capacity to adapt to climate change. The group was asked to list the different institutions that they have some links with. The linkages and networks among institutions and how communities access the institutions was drawn and discussed.

Cause-effect analysis

The group was asked to identify problems that they face in relation to the environment and rank them. Using one most important problem, they analyzed its causes and effects. The communities' responses to the problems were discussed including conflicts and governance systems.

Seasonal calendar

This tool identified sources of livelihood for the people in the villages. The sources were mapped across a full year in terms of productivity and accessibility. The group brainstormed on the opportunities for improving access and productivity to ensure adequate sources of livelihoods across the year.

2.4 Quantitative study.

The quantitative study was conducted by administering a semi-structured questionnaire as shown in appendix 3. The study interviewed about 900 respondents in Zomba, Machinga and Phalombe. The respondents were households headed by both males and females. The questionnaires were administered by a team of ten research assistants (appendix 4) who were trained on the questionnaire before the field survey. Additional key informant interviews with some members of the communities were conducted.

Table 1: Distribution of respondents

District	Hotspot Name	Traditional Authority (TA)	Name of Village(s)	Sampled households
Zomba	Lake Chilwa Zomba side and areas including Chisi Island, Mchenga and Kachulu Beaches	Mwambo	Likapa (Kachulu)	100
		Mwambo	Kotamo (Chisi)	50
Zomba	Zomba Mountain Forest Reserve, Likangala River and areas including Zilindo and Namadidi settlements	Mlumbe	Mtiya II	100
	Malosa Forest Reserve and Domasi River	Malemia	Mbuliwa Mbawa William Chilasanje	100
			Kwenje	30
	Ngwerero EPA, Mayaka Section, Steven Section and Chimbeta Section	Ngwerero	Masale I	50
Phalombe	Michesi Forest reserve and Phalombe river Bank	Kaduya	Nalingula II	100
		Nazombe	Mangulenje	40
			Swang’oma	50
	Mpoto Lagoon and Sombani River and areas including Swang'oma beach and Njalo Island	Chiwalo	Mulambe	40
	Mikongoloni Hill and Chisengeleni River	Jenala	Mumbo	30
	Chitekesa Nambazo Stretch	Jenala	Chimombo	30
Machinga	Nacala Corridor, Namanja beach and Mpiri River Catchments	Kayinga	Ningwa, Jiramu, Mpiranjala	50
		Mlombwa	Mpalume	30
	Chikala Hills, Mposa Beach, Lingoni and Zumulu River Catchments	Mposa	Mapira	150
		Chamba	Mitawa	50

2.5 Data analysis

A Statistical Package for Social Scientists (SPSS) was used for the analysis of the livelihoods in the context of their vulnerability and capital assets, the available policies, institutions and processes, prevailing livelihoods strategies and outcomes/aspirations led to identification of appropriate entry points for future interventions in the communities to reduce poverty/vulnerability**.**

Data entry and cleaning was done by two research assistants (appendix 4). The variables that were open ended were coded and changes made where necessary for easy data analysis. For any inconsistencies, variables were cross-tabulated with inconsistencies against question numbers. The questionnaires identified to have problems were cross-checked to correct the details.

3. ASSESSMENT OF THE LAKE CHILWA BASIN COMMUNITIES' LIVELIHOODS

The analysis of the livelihoods of the Lake Chilwa Basin communities highlighted the following issues.

3.1 Weak livelihoods capital assets

The Lake Chilwa Basin can generally be characterized by weak capital assets, particularly natural, financial, human and physical resources. Only social capital can be considered as strong in some cases and it is possible to build on it to improve the other assets and reduce vulnerability.

3.1.1 Human capital

Human capital simply refers to "workforce". It is a pool of competences, knowledge and skills that work to produce goods and services of economic value. These competencies and knowledge are gained through education and experience.

The Lake Chilwa Basin is highly populated with 164 people living per square kilometer. The study found out that households comprise of 5 people on average. However, about 31% of the households have more than the average household size (Table 3) and there is no difference in the number of males and females per household.

Table 2: Household sizes

Household size	Percentage households
1-3	32.1
4-5	36.8
6-10	30.8
Above 10	0.3

The people in the basin are characterized by low literacy levels with about 28% not knowing how to read and write; 14% have not attended any formal education and about 16% percent attaining at least secondary education (Table 4). Only 3.7% of the respondents had some professional skills in bricklaying, carpentry, sewing, boat and repairing.

Human capital is essential in our knowledge–based economy where knowledge is central in economic development. Knowledge is increasingly being codified and transmitted through computers and communication networks. The weak human capital in the basin does not only weaken the development capacity but also directly weakens the capacity to diversify livelihood options.

Table 3: Literacy levels for Lake Chilwa Basin Communities

Education Level	Percentage
No Education	14.4
Junior primary (std 1-5)	38.6
Senior primary (std 5-8)	32.3
Junior secondary (Forms 1-2)	8.0
Senior secondary (Forms 3-4)	5.7
Tertiary	1.0

3.1.2 Social capital

Social capital is very complex in its conceptualization and operationalization. However, the common theme among all these is the focus on social relations that have productive benefits. Social capital is about the value of social networks, bonding similar people and bridging between diverse people, with norms of reciprocity (Dekker and Uslaner 2001; Uslaner 2001). Alessandrini (2002) has also extended social capital to include voluntary associations and organizations outside the market and state. These are private organizations that are formed and sustained by groups of people acting voluntarily and without seeking personal profit to provide benefits for themselves or for others.In this study, social capital encompassed the local networks including institutional networks, benefits of the institutions to the community and access rights to livelihoods sources (governance).

There is high informal social network at village level. Many households (50%), irrespective of income level, do work together in each others´ farms by providing necessary labor for garden preparation, weeding and harvesting. In difficult times and festivals such as hungry seasons, funerals, weddings and initiation ceremonies households help each other in terms of food supplies as evidenced by 37% of the respondent households that either give or receive food support to or from relatives within the village.. However, cash reciprocities are relatively unpopular with only 25% of households involved within the village. These reciprocities also go beyond the village boundaries; about 44% do help each other in relationships involving individuals or households of different villages. For example when a person dies in a village, households withion and outside the village would gather flour and money to support the bereaved families. There is again high level of trust at household cluster level where sharing is a daily norm. For example, in the dry season, goats are left free to graze in anybody's field without prior arrangements for permission to do so. This social order has not significantly changed over the last five years as evidence by 63% of the respondents.

The basin has many migrants coming from as far as Kasungu to the lake to fish or trade in fish and rice. These migrants and the local communities have coexisted for a long time as long as the migrants are registered with the BVC chairperson, or the village head. It was observed that most of the fishing is done by migrants while the locals are involved in fish trading. During lean fishing season, some migrants return to their original villages while others remain. The remaining migrants rent fields for rice and/or maize production.

There are many social groups that communities (over 98%) belong to (Table 5). The most important among these groups are religious groups (Churches and Mosques) where 81% of the households belong. These groups are important in safeguarding livelihoods in times of need especially by providing safety nets and security. Apart from providing food support during hunger and disaster periods religion has an overall role of shaping the society. Religion and society are intricately interconnected and codependent. Religious beliefs inevitably affect areas such as politics, economics, and cultural values Politics was rated second with almost 36% of the respondents supporting political parties. The main reason given for this was the provision of security and prioritization of government services especially those supporting ruling party. This may suggest that politics brings some elements of imbalance in getting government services. People that join community functional groups (CBOs, School committees, VNRMCs etc) do so because they are satisfied in providing services to improve lives of others and feel happy when they are recognized in doing these services. It was observed that people of this category are basically well-off than others. This may confirm the Maslow's Hierarchy of Needs where after satisfying basic needs, people may want to satisfy other higher needs hence the need for self-actualization.

Table 4: Community membership to social groups

Social Group	**Percentage**
Religious (Church or Mosque)	81
CBOs	7
Political parties	36
NGO Clubs and Associations	13
Governance committees	12
Community committees	21

On the other hand, all the communities studied have a political structure with: (i) the Traditional Authority (TA), as overall head of all land issues and acts as the clan chief; (ii) the Group Village Head (GVH) who acts on behalf of the TA but controlling a number of villages; and finally (iii) the Village Head (VH) responsible for a village that comprise of a number of households. However, the development agenda is overseen by two main committees namely Area Development Committee (ADC) at TA level and Village Development Committee (VDC) at GVH level. There are also different technical committees many of which are commodity/sector based and these include: Village Natural Resource Management Committee (VNRMC), Beach Village Committee (BVC), Violence against Women Committee, Village Credit Committee, Community policing committee and many other small committees developed as need arises.

The major problem associated with these technical and development committees is weak functional organization as a result of weak motivation in community affairs by some members, and lack of common goal. It was noted that most of these committees are donor driven as such there is a perception by many including traditional leaders that the communities get monetary benefits from their donors. At Mposa, local leaders complained of losing some of their powers due the proliferation of these commodity committees especially VNRMCs and BVCs. Kayambazinthu (1999) also noted the same in his analysis of local power in the colonial and post-colonial era where the introduction of formal political institutions during colonial and post-colonial era are said to have weakened the traditional institutions because the new institutions carry out more or less the same duties and responsibilities of traditional leadership.

3.1.3 Natural capital

The Lake Chilwa Basin is blessed with a number of natural resources namely: forests, the lake and its wetland, and arable land. These resources are the source of livelihoods for the basin population but their exploitation cannot be considered as sustainable. Access to these resources also varies per resource.

The sections below provide the status and access for each of the resource:

3.1.3.1 Forests

The Lake Chilwa Basin has four main forest reserves namely: Chikala, Michesi, Zomba and Malosa. These reserves and many other village forests and woodlots have an important function of directly supplying such indispensable resources as fuel wood, food (honey, fruits and mushrooms), fodder, fiber and medicinal plants. The Chikala Forest in particular supplies all the firewood for fish smoking at Mposa fish landing sites. Firewood trading is thus a full business chain comprising of wholesalers and retailers (plates 1 and 2).

Plate 1: Firewood wholesale at Mposa Plate 2: Firewood retailing at Mposa

On the state of these forest reserves, it was reported by 75% of the respondents that productivity all reserves has declined over the past ten years. Malosa Forest Reserve is the worst affected. It is actually paradoxical that it is still called a forest reserve because it has lost almost all its big trees. This has not only affected biodiversity but also affected the livelihoods of the Kasonga Village community which relied on the forest products for a long time. They have now started farming by even encroaching into the reserves worsening the problem. The Zomba Forest Reserve is mostly affected by fires. The communities around Mtiya Village complained of government’s privatization as the cause of the reserve’s degradation.

Except for community and personal woodlots which are privately owned, access to forest reserves is divisive in relation to products being accessed. For example, mushrooms, wild fruits and birds have open access while firewood and construction poles would require a licence which is mostly ignored. Firewood is the main product accessed by households and its demand is linear throughout the year (Figure 2).

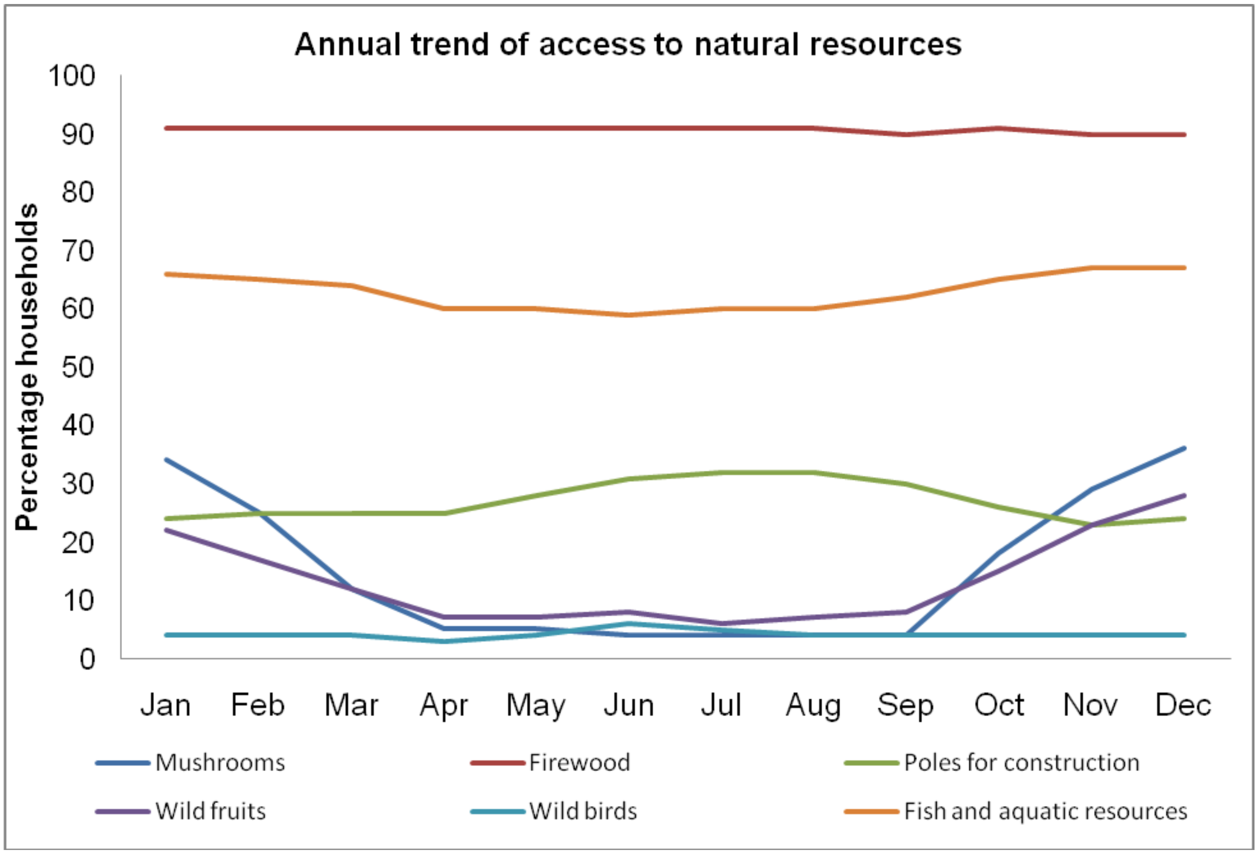


Figure 2: The trend of access to natural resources by households in a year

The management and protection of forest reserves is in the mandate of the Department of Forestry. The Department of Forestry has regulations stipulated in the Forestry Act that aim to promote sustainable forest management. Some of these include regulations on access through confiscations of illegally obtained forest products like charcoal, firewood and timber; licensing of forest users especially in plantations and issue of permits for accessing reserves by local communities. However, the success of these regulations has been sub-optimal due to many factors mainly from low forest staff numbers and financial resources.

The department is decentralizing management of these forests by involving the community in what is known as “co-management arrangements”. Literature shows that a shift towards decentralization is in recognition that an overly centralized approach to development has failed (Peters, 2000; Smoke, 2003). It is argued that downward accountability or representative authorities with meaningful discretionary powers are basic institutional elements that should lead to efficient, equitable and good forms of local governance and development (Mehta, 2000;

Ribot, 2003; Smoke, 2003). Two blocks have co-management plans in Chikala Forest Reserves. However already, members of forestry frontline staff at Mposa are complaining that local leaders are selling wood and thus the co-management system is not working well to protect the resources. Reciprocally, the local leaders complain of the staff being involved in corruption with firewood wholesalers. What was surprising was to see firewood from fresh trees being sold at Mposa landing site but nobody seemed to care. Truck and bicycle loads of firewood are ferried every day reportedly from the two co-management blocks of the Chikala Forest Reserve to the landing site. Retailing of the firewood preferably hardwood is done by women at the landing site.

3.1.3.2 The lake

Lake Chilwa is the most productive and resilient lake in Malawi. Its fishery has recovered from eleven dry ups since it was first recorded (GoM 2000a). Fish resources are the main natural capital of the lower stream of the Lake Chilwa Basin accessed without restriction by over 66% of the respondents. However, women are mostly involved in processing of fish and small ancillary businesses such as selling firewood to fish processors and food supplies. Van Zweiten and Njaya (2003) estimate that the Lake contributes between 16 percent and 43 percent, with an average of 22 percent, to the total fish catch of Malawi contributing about US$17 million a year (Schuijt,1999). The main fish species caught are *matemba (Barbus spp), Mlamba* (cat fish), Makumba (*Oreochromis shiranus*). Fishing in the lake is done by people across the country. There are migrants from different places of Malawi that come to fish in the lake. The fish is also sold to distant markets of Malawi as far as Kasungu in the central Region of Malawi.

There is a general agreement by the respondents (although 10% of the respondends felt production has increased) that fish catches have declined over the past ten years because of two reasons:

1. The lake size especially in the dry season reduces as such fish are concentrated in fewer deeper areas that also result in a large number of fishers in a relatively smaller area. The concentration of fish and fishermen results in catching even juveniles. As a result the amount and sizes caught by individual fishers decline (See Box 1).
2. Increasing numbers of fishermen mainly those using seine nets. Lake Chilwa is dominated by small-scale fishing. This sub-sector has open access to the fisheries resources in the lake except for some restrictions that have been imposed on gear types, mesh sizes and closed seasons. It is provides employment to many people

including young men. It is also viewed as a cushion to food insecurity and poverty. Therefore in times of food insecurity many people migrate to the lake to fish or get employment.

Box 1: Communities perception on fish catches at Mposa Fish Landing Site

"The fish catches have been declining over the past ten years and now it's worse …"fish is gone now and we are just like gleaning…those with seine nets have finished up the fish because they just catch anything else in the water…and there is nothing we can do…because you cannot stop your friend from making money even if he is destroying, you don't have the powers to stop him … its difficult for fish to multiply because when it goes this way it ends up in miyono (traps), when it goes there, it is seine nets (khoka), this way there are gill nets… so it is being hunted everywhere and hence the fish catch has gone down.…" W*hen the fish moves from here (the marshes where it breeds and grows) to Mpale (open waters)…it does not come back here (for breeding again) it is just caught right there at Mpale with seine nets…*

…in the past years there were a lot of big fish. We used to catch Chambo as big as 3 inches…but now…eee! You can't find it…now what is common is the 2 inches…that's what we call big fish now".

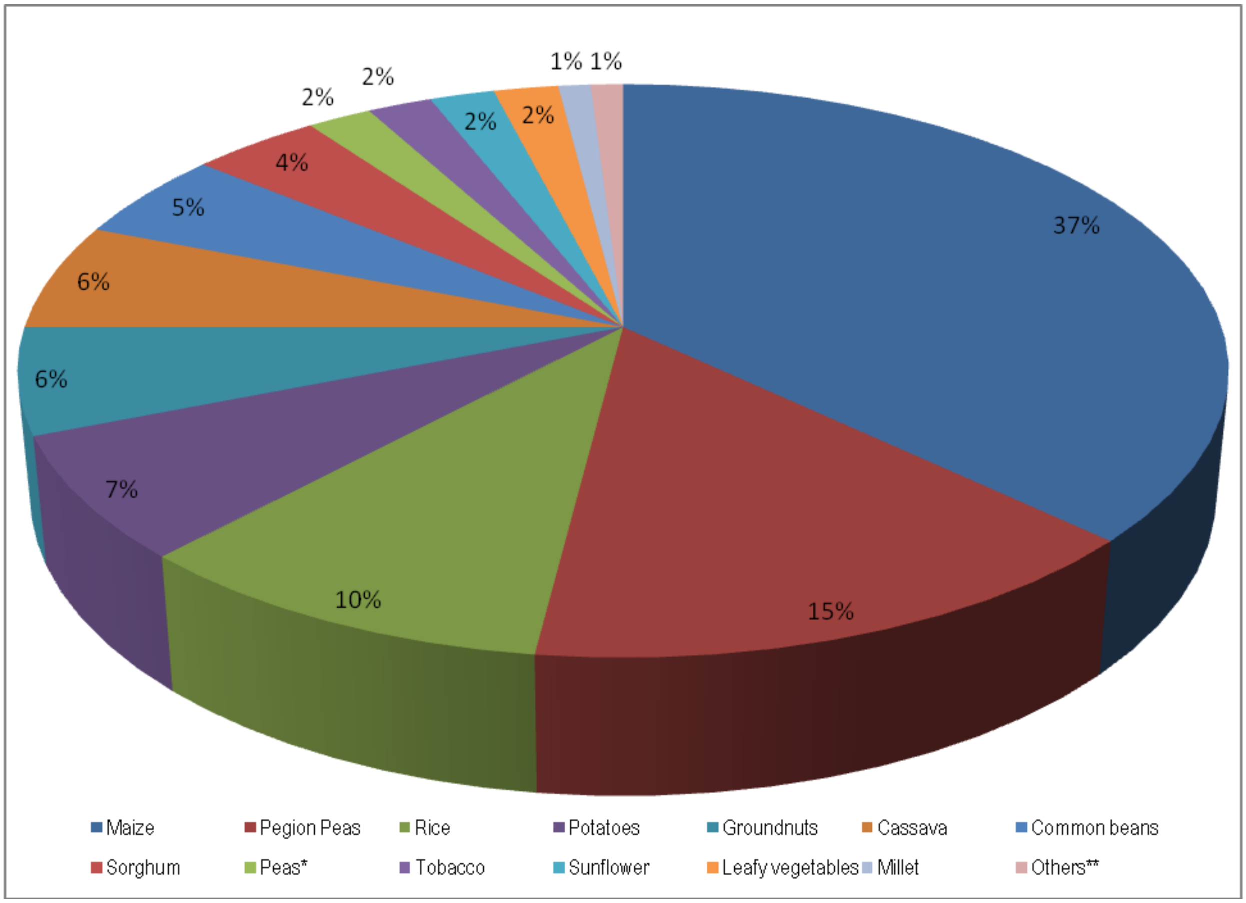
The fisheries management dates back as far as 1930 when the first government centered regulations were introduced. In line with decentralization policy, by mid 1990s, there was a gradual shift in the fisheries management philosophy from the conservation paradigm to the social/community paradigm, i.e. focusing on fisher community involvement in fisheries management. As such, the Fisheries Act of 1973 was reviewed with the intention of including community participation in fisheries management. A new act known as Fisheries Conservation and Management Act, 1997, was then passed in parliament in October, 1997 to replace the 1973 Fisheries Act (GoM 1997). Co-management was adopted in Malawi as an alternative fisheries management arrangement in view of the unsatisfactory traditional and government-centered management regimes as manifested by the general decline of the fisheries resources especially in Lake Malombe (Hara, 1998).

Following this decision, department of Fisheries has facilitated the establishment of BVCs to manage fishery resources. There are many BVCs almost on each fish landing site, however many of these BVCs are not trained and do not perform their anticipated duties effectively. BVCs have been developed without proper guidelines on institutional arrangements and training, and some of these institutions operate without a constitution and funds for their running costs. Apart from BVCs the Lake Chilwa fishers have also established small villages of floating houses (*zimbowera)*. The location chairman among other things is involved in maintaining peace and order by making sure that rules and regulations are followed. *"Here we have chairmen; they are like our village headmen. The chairman at the beach is like our TA since we report everything to him and all people coming to the lake are registered with him…"* said one of the respondents at Namanja Beach.

3.1.3.3 Agricultural land

Lake Chilwa Basin provides good land for agriculture. There are over 20 crops being grown in the basin. The main crops grown are presented in Figure 3. However the importance for each crop is different for different areas except maize. For example pigeon peas are economically more important in the middle basin while rice is very important in the lower basin. For the districts, sorghum is a very important food crop in Machinga (second to maize) as compared to cassava in Zomba and Phalombe. Similarly in Phalombe, sunflower and chick peas are very common than in the other two districts.

It was observed, however, that due to population pressure more land is being segmented and increased cultivations on steep slopes. Cultivated area in sub-Saharan Africa has more than doubled since 1961 to accommodate rapid population growth (FAO, 2007). Cultivation on riverbanks especially for beans and horticultural crops was common. Cultivation on steep slopes and river banks was observed without any application of biological or physical soil conservation measures implying high potential for land degradation. There are also numerous patches of land being irrigated by smallholders using treadle pumps, river diversions and watering cans. There are three main rice schemes in the basin namely: Domasi, Likangala and Zumulu. There are also many small irrigation schemes supported by various NGOs.



*Peas include: green peas, cowpeas, chick peas
**Others include: sugarcane, fruits, cotton, and okra

Figure 3: Major crops of the Lake Chilwa Basin

Livestock are also important. Every household studied had chickens (Figure 4) mostly for home consumption but others said the chicken are also sold when in need of cash. The lower basin populations in all the districts own cattle while pig populations are high in Phalombe.

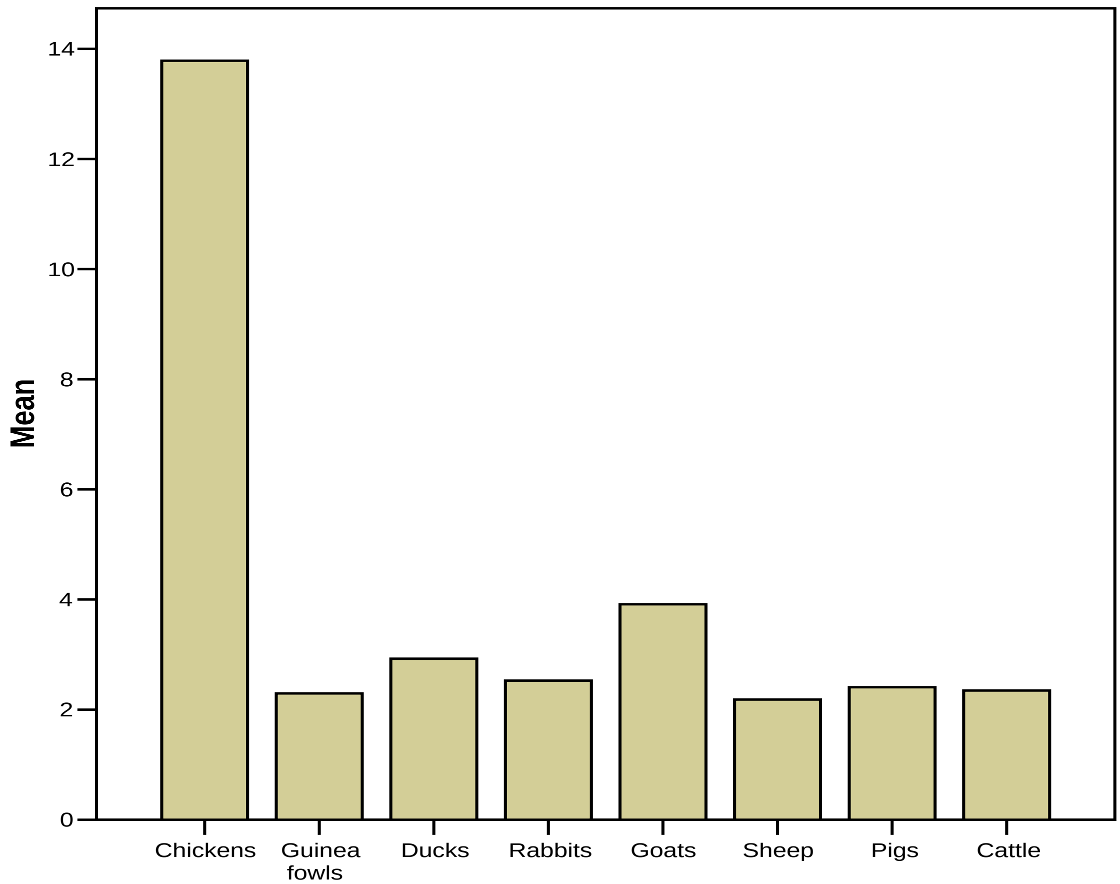


Figure 4: Livestock ownership in the Lake Chilwa Basin

Land productivity is said to have remained the same over the last ten years. This could be a consuetude perception due to the fertilizer subsidy program that has seen agricultural production going up and not necessarily that the soils are becoming rich. Other areas such as Mulambe and Ngwerero have experienced low rainfalls over the years hence affecting their agricultural production.

The people in the Lake Chilwa Basin, as is in most parts of southern Malawi, are traditionally matrilineal and have uxorilocal residence whereby husbands come to live in their wives' villages. Their inheritance of land is through female heirs as such land is inherited through the mother line except for the rice schemes that are managed by Water Users Associations (WUA). Access to land in these schemes is through applications to the WUA who then screen the applications and successful ones are asked to pay an annual subscription fee. The study also found out that many migrants have access to land especially the wetlands for rice production. This is done through a renting process by either cash or bags of rice after harvesting. Access through this system operates in social systems that are dominated by power and structures of domination (Kambewa 2008). Thus, social systems create conditions in which access to assets, income or other means of production is differentially distributed among individuals in the society (Blaikie, *et al*. 1994).

3.1.4 Financial capital

As indicated earlier, the main occupation of Lake Chilwa Basin communities is farming and this is where 37% of the households have as their main source of income (Figure 5). Other sources of income include agriculture, wage labour, fishing and fish related businesses.

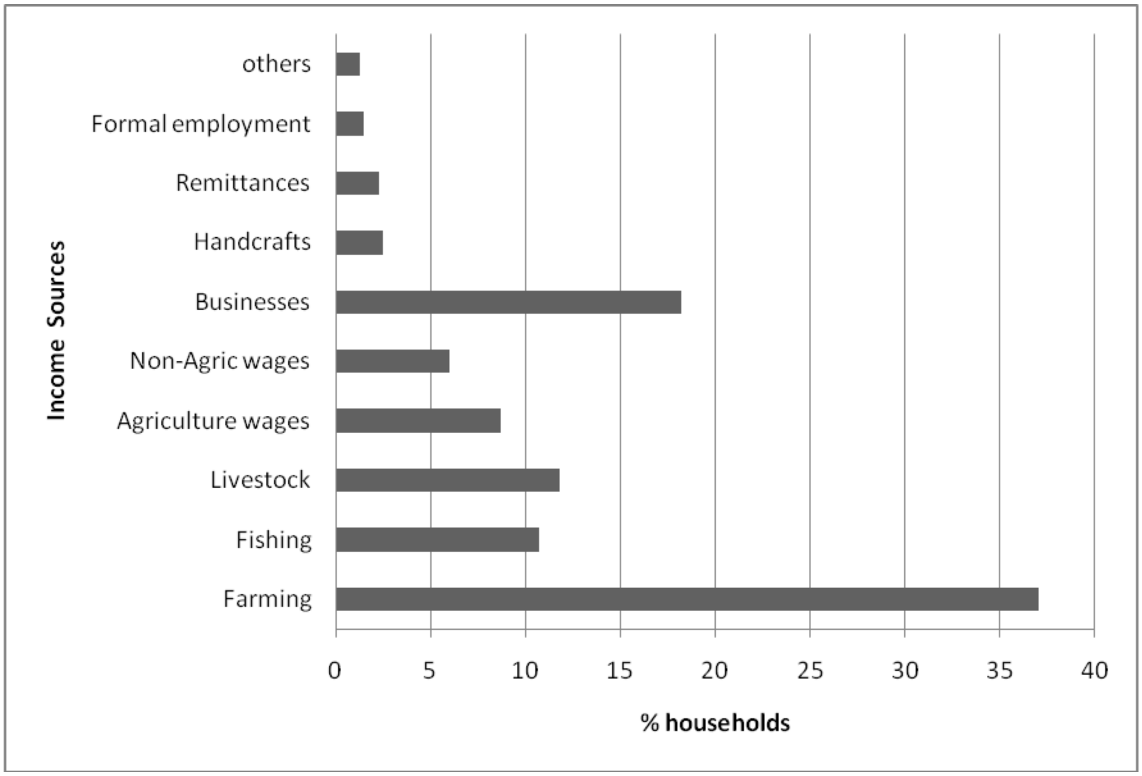


Figure 5: Sources of income in the Lake Chilwa Basin

All the sources are natural-resource based and as such they are easily affected by climate change. It implies that the communities' livelihoods get easily affected by climate variability such as rainfall amounts and distribution, temperatures on top of macroeconomic conditions. This was stated by 50% of the households as some of the reasons to reduced income levels apart from high cost of inputs and decline in landholding sizes. On the other hand 19% of the households indicated that their incomes improved over last 10 years mostly due to other business ventures which they undertook.

The biggest expenditure for the Lake Chilwa communities is food that comprise of about 66% of the total household expenditure with carbohydrates especially maize forming the biggest part not in cost but the number of households. Although the number of households spending on leisure and beer was very small but the value of expenditure was higher than that of food. Table 6 below highlights other important expenditures. The fact that the main income sources and expenditure items is agriculture (Figure 5 and Table 6) may suggest that there is limited off-farm diversification such that either for any food production there is a portion sold to satisfy immediate household cash needs, or that cash crop production is so high than food crop production and households use the cash generated from sale of the former to supplement food supplies. As such 64% of the households have hardly any surplus income for savings.

Table 5: Composition of expenditures in the Lake Chilwa Basin

Item	**% Expenditure**
Agricultural inputs	10.9
Construction	1.2
Clothing	3.4
Food	65.8
Groceries	9.1
Education	0.4
Health	1.0
Transport	1.2
Investments	3.7
Others	0.3

There are no credit facilities in the communities except for those found in urban centers of Liwonde, Zomba and Phalombe. Apart from higher interest rates that these micro-credit organizations charge, people have to walk long distances to access them. Women especially in fish landing sites have therefore promoted the development of traditional savings and credit systems known as *"Banki ya mmudzi* (Rural bank). However the proportion of beneficiaries of these credit schemes is still low.

3.1.5 Physical capital

The Basin's infrastructural capital is in dilapidated state. Schools and hospitals are widely dispersed or poorly equipped with some isolated from the main communities. Table 7 below highlights the distance to the nearest facility in the basin.

Table 6: Distance to social services

Facility	Average travelling distance (Kms)
Main hospital	26
Health center/clinic	5
Primary school	2
Secondary school	7
Major markets	4

There are generally poor road conditions which affect the distribution and marketing system of major products and inputs in the basin. Of the 3 districts, Phalombe has the poorest road infrastructure with most roads being in poor condition.

The artisanal fish landing sites are devoid of basic fisheries infrastructure, such as fish handling and processing areas, storage facilities for processed products. One of the major challenges for fish marketing is lack of cold storage facilities and ice production. For example most fishers go out fishing without ice and degradation of fish sets in immediately after the catch if not put on ice. With high fish landings coinciding with rainy season and high humidity, the impact of these shortcomings is quite significant. It is estimated by the Department of Fisheries that the post harvest losses incurred by fishers and traders could be as high as 50% especially for fresh fish. In some cases fishers tend to reduce prices in order to dispose off the fish quickly to avoid spoilage.

3.2 Weak Policy Framework

Despite the available policies dealing with land, water, forestry and fisheries resources, Malawi still experiences decline in natural resources due to overexploitation and environmental degradation, which is linked to gaps and inadequacies of available policies, legal and institutional frameworks. Specific major issues include inadequate policy and legal framework, weak institutional framework for policy implementation, poor coordination, poor monitoring and evaluation, weak policy implementation and life span of polices. The government ministries through various departments implement the policies however enforcement is not prioritized. It is often not given enough cash and human resources as such its highly corrupted. Some of these national policies on natural resources include the following:

3.2.1 The Malawi National Land Policy (2002)

The policy seeks to ensure tenure security and equitable access to land, to facilitate the attainment of social harmony and broad based social and economic development through optimum and ecologically balanced use of land and land based resources.

3.2.2 The National Land Resources Management Policy and Strategy (2000)

As agriculture is the biggest land use in Malawi, the National Land Resources Management Policy and Strategy goal is to promote efficient, diversified and sustainable use of land based resources both for agriculture and other uses in order to avoid sectoral land use conflicts and ensure sustainable socio-economic development.

3.2.3 The National Water Policy (2005)

The policy as revised in 2005 is meant to address all aspects of water including resource management, development and service delivery. The policy covers areas of water resource management and development, water quality and pollution control, water utilization, disaster management and institutional roles and linkages. It also promotes efficient and effective utilization, conservation and protection of water resources for sustainable agriculture and irrigation, fisheries, navigation, eco-tourism, forestry, hydropower and disaster management and environmental protection.

3.2.4 The National Irrigation policy and Development Strategy (2000)

The policy's specific objective is to manage and develop water and land resources for diversified, economically sound and sustainable irrigation and drainage systems under organized small-holder and estate management institutions and to maintain an effective advisory service.

3.2.5 The Agriculture Policy Framework 2005:

The objective of this framework is to harmonize national agricultural development strategies and agricultural-related legislation and policies

3.2.6 The National Fisheries and Aquaculture Policy (2001)

To improve the efficiency of all aspects of the national fisheries industry, the production and supply of existing fisheries products, as well as development of new products to satisfy local demands and potential export markets. The policy provides strategies to ensure sustainable exploitation of the fisheries resources without undermining the environment.

3.2.7 The National Forestry Policy (1996)

In recognition of the important role the forestry sector plays in the national economy and the livelihoods of the poor, the policy was developed with legislation enacted by Parliament in 1997. The goal of the policy is to sustain the contribution of the national forest resources to the improvement of quality of life in the country by conserving the resources for the benefit of the nation.

3.2.8 The Environment Management Act (1996)

The policy currently being revised, seeks to harmonize all environmental acts such as the Forestry, Fisheries, Wildlife, Water Resources, Irrigation and Land Resources Conservation to allow for an enabling environment for the development of co-management systems for natural resources and for fostering partnerships between rural communities, Government, NGOs and the private sector

In addition to these legal instruments, the Government of Malawi is a party to many internationals agreements and regional protocols that she has to conform to. These agreements and protocols relevant to natural resources conservation, management and development include: the Convention on Wetlands of International Importance (Ramsar Convention), the Protection of the World Cultural and Natural Heritage Convention, The International Trade in Endangered Species Convention, Biological Diversity Convention and the Cartagena Protocol on Biodiversity, Food and Agricultural Organization (FAO) Code of Conduct for Responsible Fisheries and Southern African Development Community Protocol on fisheries.

The analysis also reveals weak links between the micro-level (communities) and meso/macro-levels (national, international). Communities are not aware of the national institutional and political framework, including regulations and the opportunities to improve their livelihoods at the local level. Communities have an indirect say in the policies that are meant to improve their living conditions. Their views are expressed to development partners and agencies through their leaders (mostly politicians). Important discussions of national importance are organized with community leadership who in turn consult with members of the community. Consequently development policies are designed and implemented without effective participation of the people concerned.

3.3 Vulnerability of the Lake Chilwa Basin Communities

Four main factors make the basin communities vulnerable:

3.3.1 Seasonality of livelihoods sources:

The main livelihoods sources – farming, fishing even businesses are seasonal. The households have enough income from farming especially rice and fishing during the period June to September. It was particularly pointed out that fish trading is highest in June (Figure 6) because this is the period that they have high *matemba* (*Barbus spp*) catches and high incomes from farm products.

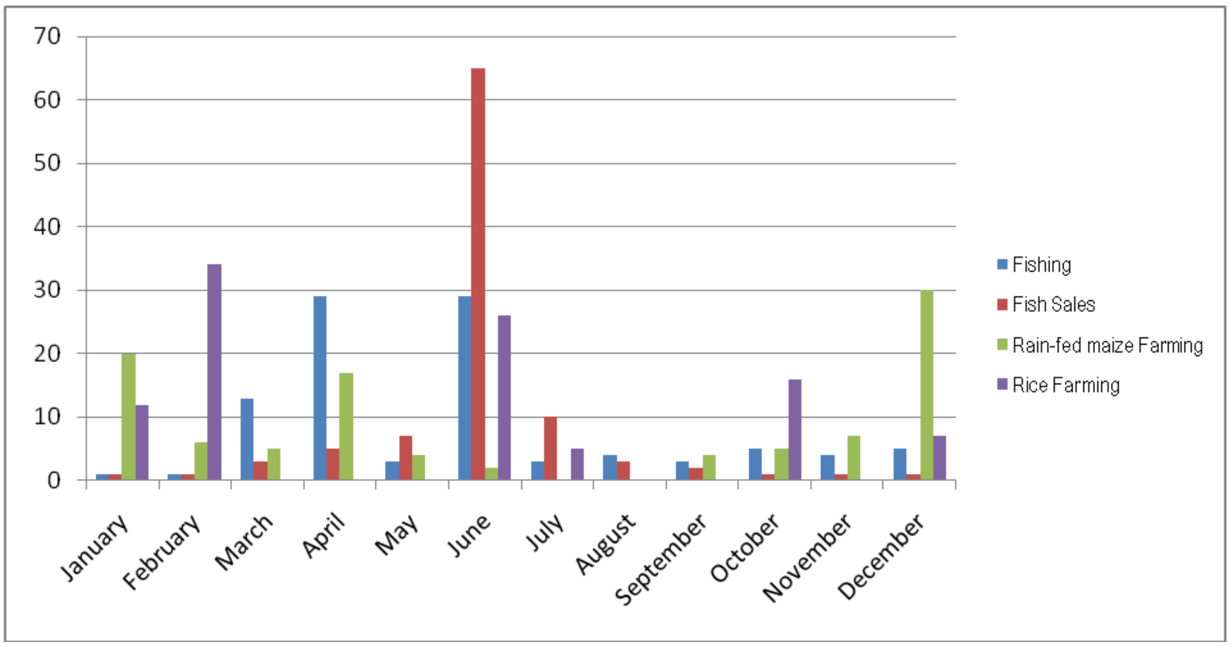


Figure 6: Seasonal calendar for major livelihoods sources in the basin

Households have lowest income levels from January to February. This the period when the lake is closed for active fishing therefore low incomes from fish, furthermore this is a period with high farm labor demands especially weeding in maize fields and transplanting of rice requiring high cash demand. Conversely, this is also a period when 42% of the households have food shortage (Figure 7). This condition forces some households to migrate to urban centers or sale labor (ganyu) to richer households for food supplements which in a longer run creates a vicious cycle of household poverty.

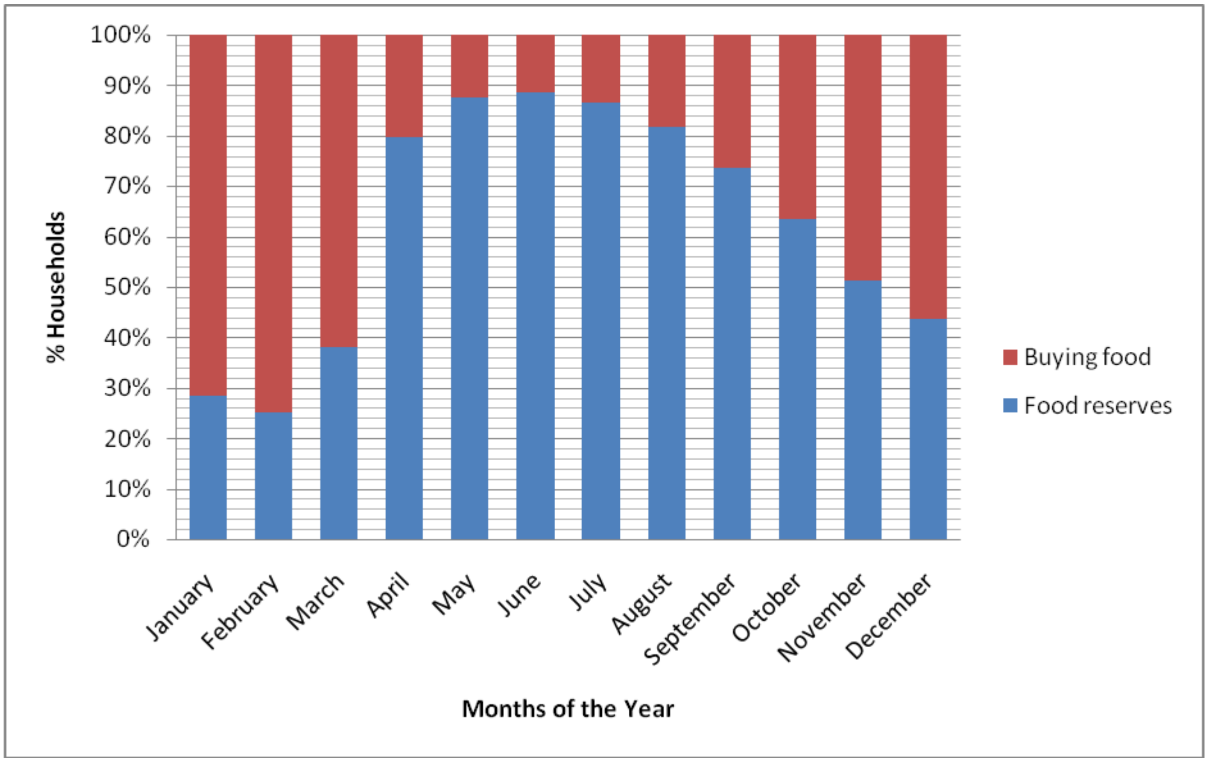


Figure 7: Household food accessibility

3.3.2 Low off-farm livelihood diversity coupled with over-dependency on natural resources

Households in the basin have on average only two major sources of incomes: farming and natural resource-based small businesses Farming is the main source of income (Figure 5). There are many small businesses on farm and natural products such as trading on fish, rice, firewood, making and selling of mats made from reads and marrshes. With this limitation in off-farm livelihoods diversification households' resilience to climate variability is very weak as evidenced during years of even small droughts.

3.3.3 Existence of three preventable diseases affecting the community (malaria, diarrhea and bilharzia).

Illness of atleast one member of the household is the major shock in the Lake Chilwa Basin experienced by over 56% of the households. Malaria is the major disease whose prevalence stems from two factors: the presence of swamps and misuse of bed nets. Diarrhea is caused by the lack of safe and clean drinking water in most communities especially in the fish landing sites. Potable water supply systems and sanitary facilities in fishing communities in particular are not available and consequently environmental hygiene and beach sanitation are major problems. There are many fishers that stay in floating huts (*Zimbowela*) on the lake marshes. These fishers sleep, eat and bathe out in the marshes. They drink the water from one side of the floating hut and use the other side for waste disposal including human waste. During severe cholera outbreaks, 40 – 45 percent of these inhabitants are affected.

3.3.4 Weak institutional support and framework to guarantee sustainable natural resource management and development:

Policies in sectors such as water, agriculture, health, infrastructure, and economic development can facilitate or constrain adaptation. Over the last decade the concepts, policies and practices of conservation in Africa have begun to shift towards what has been viewed as a community-based approach. Although these challenges have commonly been perceived as a move towards putting `community in conservation' (Agrawal, 1997) and are widely referred to as community-based conservation (Western and Wright, 1994) or participatory approaches, they are more complex than the simple dynamic of a shift of responsibility and authority from the state to the community suggests. Communities' participation in natural resource management is passive, consultative and functional. Therefore, the participation process should analyze relevant policies, focusing on the integration of climate change issues into policies, and on opening barriers to facilitate adaptation in the communities. The box below presents different participation styles and their consequences.

Box 2: Types of community participation

Passive Participation: *People participate by being told what has been decided or has already happened. Information being shared belongs only to external professionals.*

Participation by Consultation: *People participate by being consulted or by answering questions. Process does not concede any share in decision-making, and professionals are under no obligation to take on board people's views.*

Bought Participation: *People participate in return for food, cash or other material incentives. Local people have no stake in prolonging technologies or practices when the incentives end.*

Functional Participation*: Participation seen by external agencies as a means to achieve their goals, especially reduced costs. People participate by forming groups to meet predetermined objectives.*

Interactive Participation*: People participate in joint analysis, development of action plans and formation or strengthening of local groups or institutions. Learning methodologies used to seek multiple perspectives, and groups determine how available resources are used.*

Self-Mobilisation and Connectedness: *People participate by taking initiatives independently to change systems. They develop contacts with external institutions for resources and technical advice they need. but retain control over how resources are used.*

Source: Pretty, Jules, 1995.

There are various organizations that work in the communities (Figure 8). These organizations are in two levels: those that are in direct contact with the community (represented by the inner grey circle) such as CBOs, faith based groups and local resource management groups such as VNRMCs and BVCs. The other level involves district and national organizations dealing with policy and other social services (represented by the outer blue circle) such as government departments and the district councils. It was observed that many of the government departments work through local committees. However the working relationship between them is disjointed and the coherence of messages by different sectors is often lacking such that one community member can seat in more than three different committees in the same week. On the analysis of reliability and trustworthiness of the organizations, the government ranked very low on reliability while CBOs and NGO ranked very low on trustworthiness *"….government programs take time to be implemented… NGOs' and CBOs' choice of beneficiaries is always biased and only few people in the village benefit, besides we even don't know their budgets*," a comment by participants at Mapira Village FGD.

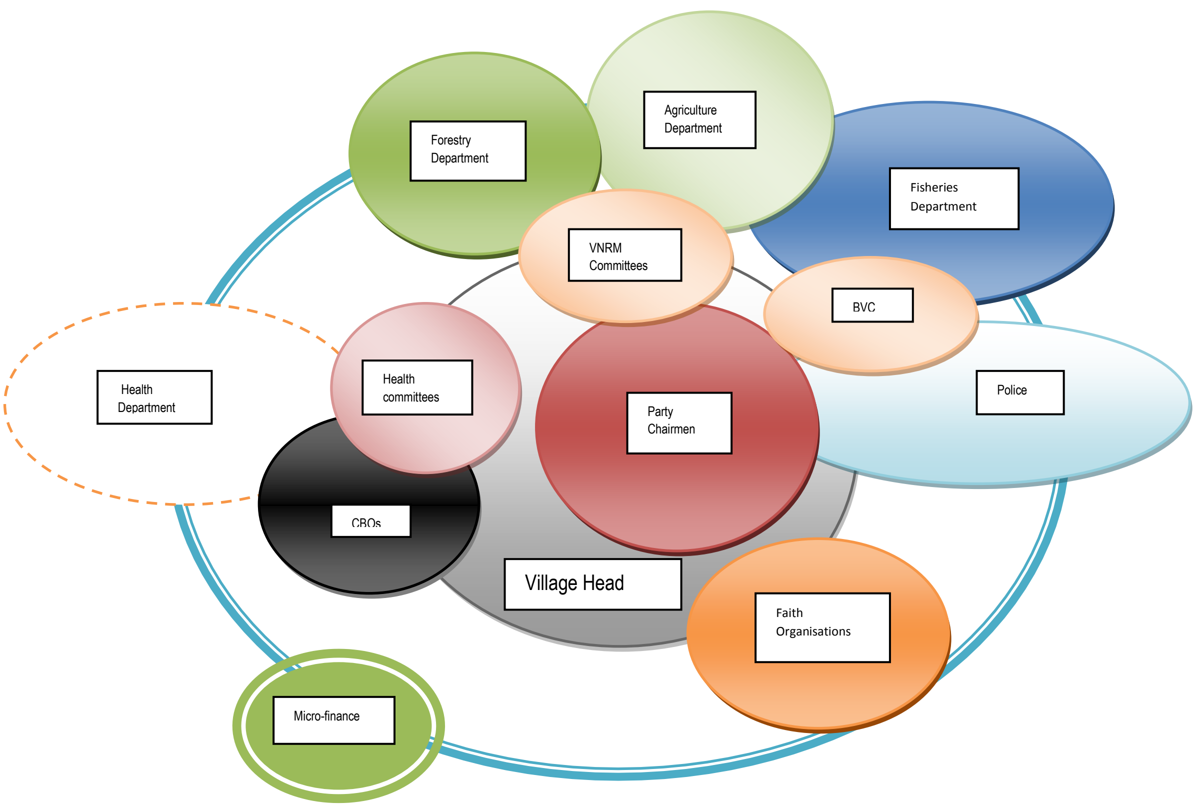


Figure 8: Venn diagram of Lake Chilwa Basin

3.4 Livelihoods coping strategies and Outcomes

The consequence of weak capital assets and unfavorable institutional environment are the causes of the limited opportunities available for livelihoods diversification and development for communities to reduce vulnerabilities. A livelihood is resilient when it can cope with stresses and shocks. Coping is here defined as ‘short-term responses’ to unplanned crises. In extreme food insecurity months, 23% of the households are involved in selling labor with women taking the biggest brunt. Men migrate to the lake to look for employment while others (2%) migrate to nearby cities and towns to engage in odd jobs (casual laborers).Livestock are also very important in cushioning 14% of the households to shocks especially during famine and other cash demands especially during illnesses. Other coping strategies are presented in Table 7.

Table 7: Major coping strategies to common vulnerabilities in the Lake Chilwa basin

Coping strategy	% households involved
Short term loans from kins	9
Food aid from kins	17
Government relief	4
NGO relief	2
Reduced meal frequencies during famine	19
Selling of household assets	5
Livestock selling	14
Selling of family labor (ganyu)	23
Migration	2
Irrigation	2
Selling other grains (rice) to buy maize	9

From the table above, it is noted that many of the coping strategies only puts the households in a vicious cycle of vulnerabilities. For example, ganyu, migration and reducing meal frequencies may result in reduced labor for household own enterprises consequently their capacity to recover when conditions become favorable is slow or limited; similarly loans, selling livestock and household assets further reduces their economic capacity to deal with production demands such as cost of inputs and children’s education.

Limited opportunities for off-farm livelihoods diversification and insufficient incomes generated from the activities of the basin communities impact the social life of the communities. However, households aspire to see the quality and standard of their lives improved through:

- Expanded irrigation facilities
- Access to microcredit opportunities
- Access to adequate basic socio-economic services (safe drinking water, toilet facilities, health care, education)
- Better access to fishing materials and equipment (nets, canoes, etc.);
- Construction of better and improved housing facilities.

4. CONCLUSION AND RECOMMENDATIONS

The people of the Lake Chilwa Basin face many constraints, such as high demand for natural resources, low incomes, lack of off-farm employment, low levels of literacy and education, poor health, soil erosion and declining soil fertility. Poverty and vulnerability of the basin's population is high with over 80% of the households having no food by January. The Lake Chilwa Wetland State of the Environment Report states that, by the end of February every year, 90 percent of all households in the basin are short of food. A key point is that there is scope for improving livelihoods even in contexts where natural resource dependence is high and alternative income-generating activities are possible.

The study found that illiteracy, poor access to credit and weak organizational capacities of the communities in the lake Chilwa Basin – were the major reasons behind the low participation of communities in decision-making processes, especially in forestry resources management and local development. The improvement of human and social capital of the communities and the political and institutional environment could be entry points to making physical, financial and natural capital more efficient, hence reduce vulnerability.

In order to maintain the food security, economic and social benefits of agricultural and natural resources in the face of known climate variability and anticipated climate change, cross-sectoral collaboration at local, basin and national scales is required. The following are recommendations to promoting adaptive capacity of Lake Chilwa Basin Communities for sustainable livelihoods:

a. Building social capital for empowerment

Livelihoods can be improved by addressing the institutional causes of vulnerability and insecurity. Future interventions will have to focus on building social capital, for example through development of professional organizations, particularly by improving the existing informal groups. The capacity of communities will be enhanced by making them familiar with the essentials of organizational development and leadership skills. This will help them to be real partners to government and other development institutions. Community members should be trained and supported to set up functional structures to enable them to participate effectively in the decision-making process while being able to protect their interests in natural resource governance.

b. **Building strong financial service:**

The analysis showed that it is possible to build on existing informal credit organizations in selected villages to improve access to credit or enhance savings. There is need to better organize informal credit organizations by improving their functionality and linking them to existing financial institutions, and to develop a savings culture especially during rice harvesting and good fishing seasons.

c. **Adding value to existing commodities:**

There are currently seven major commodities of economic importance namely: fish, rice, pigeon peas, cassava, sunflower and firewood. These commodities are poorly marketed. Market chain analysis of these is important in order to understand actors and value addition in order to increase incomes and employment. Firstly, linkages with traders and processors through the development of agribusiness centers can immediately improve incomes. Secondly, development of small agro-processing units e.g. grading, cleaning and packaging can offset the seasonality of these products, add value hence increase producer incomes.

d. **Building resilience of communities to climate variability and change**:

Enhancing local-level adaptive capacity and building resilience of communities to climate variability and change must focus on strategies that lead to livelihoods improvements. There is need to expand the current livelihood options in the basin in order to enhance local-level adaptive capacity. For example, the Lake Chilwa Basin has good potential for irrigation development. Irrigation can be promoted with the development and promotion of drought-tolerant and early-maturing crop species. These developments need to be supported by a strong meteorological service to provide timely weather and climate information early-warning systems. Additionally, improvements to other alternative income-generating activities such as agro-processing and provision of services can expand the livelihoods base. The fishing industry currently provides cushion during hunger seasons. However the fish value chain needs to be improved to ensure that the primary stakeholders (fishers and processors) maximize profits that will create employment at the local level. Chilowa (1998) found that smallholder farmers in Malawi were losers while private traders and state marketing agencies benefited from agricultural policy reforms led by the market-led approach.

e. Strengthening the Natural resources policies' frameworks

It is vital for the project to improve ways of working to sustain future interventions in the communities. Formulation of strategies and actions as interventions for the identified policy gaps and inadequacies will lead to successful conservation, management and development of the natural resources in the Lake Chilwa Basin and the whole country.

On the other hand, there is need to balance power relationships between the state and communities. Power plays a central role in dictating and controlling access to the means of production and distribution of rights and benefits in the society. Campbell and Olson (1991) commented that "decisions regarding which resources to develop for which industries and where such activity will occur are not arbitrary, rather, they reflect the objectives of the powerful interest groups whose power is mediated through political, social and economic institutions and is expressed in the temporal and spatial patterns of resource exploitation". It is therefore recommended that bold policy decision such as a 3 years ban of tree harvesting be made to protect forestry reserves of Malosa and Zomba. However communities around these forests, need to be taught on the advantages of such a policy decision and given other income generating activities such as bee keeping. As the Brundtland Report (1987) has cautioned `it is both futile and an insult to the poor to tell them that they must remain in poverty to protect the environment'.

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APPENDICES

Appendix 1: Steps in the Livelihoods Analysis

Step 1: Define sites within the hotspots to do the analysis
Need to determine villages representing (1) upstream, (2) middle stream and (3) downstream
We therefore need 3 villages per hotspot

Step 2: Collection of Secondary data
Relevant information from reports, development plans and census data will help to build up knowledge of the hotspots.

Step 3: Assessment

Guiding Questions	Tools
– **Human Capital** – How complex is the local environment (the more complex, the greater importance of knowledge?) – From where (what sources, networks) do people access information that they feel is valuable to their livelihoods? – Which groups, if any, are excluded from accessing these sources? – Does this 'exclusion' affect the nature of information available? – Are knowledge 'managers' (e.g. teachers or core members of knowledge networks) from a particular social background affect the type of knowledge that exists in the community? – Is there a tradition of local innovation? Are technologies in use from internal or external sources? – Do people feel that they are particularly lacking in certain types of information? – How aware are people of their rights and of the policies, legislation and regulation that impact on their livelihoods? If they consider themselves to be aware, how accurate is their understanding	• Questionnaire – communities, key informants • Venn diagrams • Social mapping
Social capital – What social networks and more formalised groups (vertical and horizontal) exist? – To what extent do such networks and formalised groups build trust, facilitate cooperation and expand access to wider institutions? – How does such trust, cooperation and access influence livelihood opportunities?	• Social mapping - Actor network analysis and 'power network diagrams' • Venn diagrams
Natural capital – Which groups have access to which types of natural resources? – What is the nature of access rights (e.g. private, common?). How secure are they? Can they be defended against encroachment? – Is there evidence of significant conflict over resources? – How productive is the resource (e.g. soil fertility)? Has this been changing over time (e.g. variation in yields)? – Is there existing knowledge that can help increase the productivity of resources? – Is there much spatial variability in the quality of the resource?	• Questionnaire – villagers and key informants • FGDs • Natural resource mapping

– How is the resource affected by externalities (e.g. cross-boundary impacts of other users)? – How versatile is the resource? Can it be used for multiple purposes (this can be important in cushioning users against particular shocks)	
Physical capital – What is the state of transport infrastructure and services, ICT and energy supplies? – How do they directly or indirectly support livelihoods? – What access do people have to them? – Do people have adequate shelter, water supply and sanitation?	• Questionnaire • FGDs • Venn diagrams
Financial capital – In what form do people currently keep their savings (livestock, jewellery, cash, bank deposits etc.)? – What are the risks of these different options? How liquid are they? – Which types of financial service organisations exist (both formal and informal)? – What services do they provide, under what conditions (interest rates, collateral requirements etc.)? – Who - which groups or types of people - has access? What prevents others from gaining access? – What are the current levels of savings and loans? – How many households (and what type) have family members living away who remit money? – How is remittance income transmitted and how much is transmitted? – How reliable are remittances? Do they vary by season? – Who controls remittance income when it arrives? How is it used?	• Questionnaire – communities, key informants • Seasonal calendars
Transforming structures and processes – What policies and legislation influence people's livelihoods, and how? – What institutions and organisations influence people's livelihoods assets, strategies and outcomes, and how do they influence these? – What are the roles and responsibilities of these different institutions and organisations, and how are they established and enforced? – What are the interrelationships between these different institutions and organisations? – What access do poor people have to these institutions and organisations? – How aware are people of their basic human and political rights? How are rights enforced/safeguarded? – Are there organisations and processes that are owned by poor communities and groups, and that engage in self-help and advocacy initiatives?	• Secondary information - policies • Venn diagrams • Actor network analysis and 'power network diagrams' • Cause-effect and flow diagrams • Market inventories and commodity price tracking • Narratives or institutional histories from key informants (including on traditional rules, tenure and markets)
Livelihoods strategies – What does the livelihood 'portfolio' of different social groups look like (percentage of income from different sources, amount of time and resources devoted to each activity by different household members, etc.)? – How and why is this changing over time? (Changes may be, for example: long-term, in response to external environmental change; medium-term as part of the domestic cycle; or short-term in response to new opportunities or threats.) – How long-term is people's outlook? Are they investing in assets for the future (saving)? If so, which types of assets are a priority?	• Ranking of income sources • Mapping of migration patterns • Inventory and ranking of income and expenditure • Seasonal calendars of production, employment and income

– How ‘positive’ are the choices that people are making? (e.g. Would people migrate seasonally if there were income earning opportunities closer to home or if they were not saddled with unpayable debt? Are they ‘bonded’ in any way? Are women able to make their own choices or are they constrained by family pressure/local custom?) – Which combination of activities appear to be ‘working best’? Is there any discernible pattern of activities adopted by those who have managed to escape from poverty? – Which livelihood objectives are not achievable through current livelihood strategies?	
Livelihoods outcomes – What are the kinds of livelihood goals that people aspire to achieve and what is the relative emphasis that they place on different livelihood outcomes? (e.g. more income, increased well-being, reduced vulnerability, improved food security, more sustainable use of the natural resource base). – What trade-offs or conflicts are there between these different livelihood outcomes? – To what extent do people actually achieve their livelihood goals, and what is preventing people from fully achieving them?	• Well-being ranking of social groups, communities • Social mapping • Cause-effect diagrams
Vulnerability – Which groups are engaged in which livelihood activities? – How important is each livelihood activity to the groups that engage in it? – Is the revenue from a given livelihood activity used for a particular purpose e.g. if it is controlled by women, is it particularly important to child health or nutrition? – What proportion of output is marketed? – How do prices for different products vary throughout the year? – How predictable is seasonal price variation? – Are the price cycles of different products all correlated? – What proportion of household food needs is met by own consumption and what proportion is purchased? – At what time of the year is cash income most important? Does this coincide with the time at which most cash is available? – Do people have access to appropriate financial services to enable them to save for the future? Does access vary by social group? – To what degree are groups vulnerable to extreme climatic events and how are their livelihoods affected by such events? – How do income-earning opportunities vary throughout the year? Are they agricultural or non-farm? – How does remittance income vary throughout the year?	• Questionnaire, key informants, • FGDs • Secondary data on demography and historical market fluctuations

Appendix 2: Focus Group Discussion Guide

EXPLORING WELL-BEING AND ILL-BEING

The concept of well-being is broader than poverty which is usually considered as linked only to economic criteria. The challenge is to understand people's definition of well-being; what kinds of factors do they include in their definitions of well-being and ill-being. The following broad questions need to be explored:

- How do people define well-being or good quality of life and ill-being or bad quality of life?
- How do people perceive vulnerability, risk, security and opportunities? Have these changed over time?
- How do households cope with decline in well-being? How do these coping strategies in turn affect their lives?
- What are people's hopes and fears for the future?

Methodology:

There are different ways in which this analysis can be initiated with a group in the community. However, the following can be tried:

1. **Exploring well-being:**

In what ways do households differ from each other in this community? Or what is good/bad life? Let the group identify well-being categories.

Then go on to discuss the criteria on the basis of which households are differentiated. Pay particular attention to sources of livelihood and food security. Also ask about the types of crops that these groups grow and the type of food that they eat

2. **Scoring:**

Obtain proportions of households in each category. (The group may start by first deciding the maximum limit out of which the scores will be given e.g. 100 or 50. Avoid using smaller numbers like 10 for the maximum as it is difficult to show the proportions when there are several categories.)

3. **Trend analysis:**

The use of this method should bring out the changes that have been taking place in the community and people's perceptions about the future. The group can be asked to discuss whether there have been any changes in the well-being of the community over the last 10 years.

Have there been any changes in the number of categories (more or less of them now as compared to the past)?

Have the criteria for defining the well-being categories changed?

Obtain proportions of households in each category 10 years ago (Can also obtain proportion of female-headed households for now and 10 years go.

Have the numbers of households in each category changed (whether some households have become better or worse-off than before)? Have the sources of livelihood changed? Why? How about changes in terms of food security?

What are the factors that have brought about these changes? (Look out for issues of risk, vulnerability, opportunities, and security). Has the environment played any role in this change?

How do different groups cope with these changes?

Which livelihood strategies appear to be best for the people? What are peoples' priorities in terms of investing in assets?

Ask the group whether they perceive any changes in the situation in future and what these changes could be.

RESOURCE MAPPING

Ask the group to draw a map of their area. Tell the group that we would like to understand how life is for members of the community. Ask them to try to draw a BIG map with clear details of the important places. Give them about 15 minutes to draw it, with discussion happening as it is drawn, to make them feel comfortable. Then ask the following follow-up questions:

Which places are important for the people of this community? Do different groups have different places that are important to their lives? (Well-being groups, gender, fishers, farmers, hunters) Why are these places important?

What natural resources are available in this community now?

If you drew a map of this same area 10 years ago, would the map be same as the current one? What would be different?

What natural resources have changed? Have they changed for the better or for the worse?

What sources of livelihood have changed? Have they changed for the better or for the worse?

Are there any customs, rules and norms in this community that facilitate or hinder access to natural resources? How do they facilitate or hinder access? Were there any customs, rules and norms that used to be practiced but are no longer being practiced now?

Would this map look the same 10 years from now? What would look different?

Will things change for the better or for the worse? If things were to change for the better, what needs to be done?

What strengths and opportunities would there be in order to achieve this? How about threats and weaknesses?

INSTITUTIONAL ANALYSIS: Matrix Scoring and Ranking; Venn Diagrams

This aims to explore the most important formal, informal, government, and non-government institutions within or outside the community that influence people's lives positively or negatively. It is important that before institutional analysis is initiated with a group in the community, there is some discussion on the concept of 'institution'. The main challenge is that the term 'institution' does not easily translate into the local language. However, this should be fairly easy if this discussion comes immediately after Resource Mapping where the group has already identified institutions that support people's lives. The group can be asked to list the different institutions that they have some links with. Pay attention to how the community accesses information.

Methodology:

Ask the group the following questions:

Where do you get help from (advice, instructions, help, and support)?

Which institutions support your livelihood?

Which groups access this information and support?

1. **Scoring:**

Scoring will enable the institutional analysis to be carried out on the basis of multiple indicators. Place the identified institutions along the first column of a matrix. The matrix can be prepared on a flip chart using marker pens.

Next ask the group to discuss the basis on which they differentiate among these institutions. Allow the group to generate their own criteria. Also ask the group to consider the following criteria; trust, accessibility, effectiveness, relevance, and reliability.

Ask the group to give scores for all the institutions in the list against each of the selected criteria. Obtain reasons for every score that is given.

The following is a hypothetical example of institutional analysis using the scoring method.

Institutions	**Criteria** (scoring out of 50; the higher the score, the better the performance of the institution)			
	Trust	Effectiveness	Relevance	Reliability
Headman	30	50	40	40
Radio C	50	5	40	50
Lake Chilwa	25	30	50	10
Chikala Hills	20	30	50	15

NOTE: Do not add up the scores in the cells along the rows. This total is not a true reflection of the importance of the institutions, since the criteria do not carry equal weight.

Draw a Venn diagram to show the interaction and relationships of these institutions. The institutions can be ranked at the end.

CAUSE-EFFECT ANALYSIS

Cause-effect analysis helps to understand an issue in a more complete form. Since we are interested in understanding people's perception on access to natural resources, we can use this method to open discussions on the subject (i.e. rights, structures, and processes). Ask the group the following questions:

Are there any problems that you face in relation to the environment?

Ask the group to identify problems that they face in relation to the environment. Ask them to identify one important problem. Ask them to describe some of the causes of the problem. Place these causes on one side the 'problem'.

Similarly, ask about the effects of the 'problem' and place them on the other side of the 'problem'

Next, ask the group if there are any links between the different causes and effects. These links should be shown by arrows.

Have there been any conflicts over resources? Which groups were involved?

Over the years, have there been any changes in the rainfall pattern? How about changes in terms of temperatures?

Are there frequent dry spells now than in the past? Who gets affected? How do they cope?

What has brought about these changes? What are people doing about these changes?

SEASONAL CALENDAR

What are the sources of livelihood for the people of this community?

During which months in a year are these sources productive?

Compared to 10 years ago, has the production been changing? Why?

In terms of food availability in the home, which are the lean months? How do people cope?

During which months do people have surplus food in the home?

In a year, are there any conflicts in terms of access to resources?

How can the productivity of the sources be improved?

Appendix 3: Household Questionnaire

LAKE CHILWA BASIN CLIMATE CHANGE ADAPTATION PROGRAMME

Livelihood Analysis Household Questionnaire

Name of Household:………………………………	No. Respondents/HH……………………………….
Hotspot name:…………………….......................	Village……………………...................................
District:…………………………………………….	T/A:…………………………………………………
Date:……………………………………………….	Enumerator:…………………………………………

A) LOCATION

(Give a description of the location of the compound/house, considering :)

1. Route (*How to find the dwelling – indicate coordinates and any other features*)
   ....
   ....
2. Compound clustering: .... ***(1= <3 houses; 2= 3-10houses; 3= >10 houses)***
3. Distance to market ***(in meters/kms/miles – please indicate unit***):
   ....
4. Distance to track/road: ....
5. Distance to Health facility: Main Hospital […………..] Clinic/Health center […………]
6. Distance to School: JP School [……….] FP School [………] Secondary School [………]

B) HOUSEHOLD HISTORY

Use these questions in a flexible way to ***describe*** *the settlement and family history of the household.*

1. When and how did you start your own household?
   Year [.................................]
   How: [................] ***(1=got married willingly, 2=forced into marriage, 3=orphaned, 4=others (specify………………)***
   Where was that?[…………………………………] ***(if in present village: Go to 5; if not: Go to2)***

2. What were your main economic activities in that place? [....................]
   (1= farming, 2=fishing, 3=employment, 4=selling labour/ganyu, 5=others (specify…………...)
3. When did you leave that place? [.........................] ***year***
4. Why did you leave that place? [.........................] ***1= search for better soils, 2=search for enough land, 3=changed economic activities, 4=employment transfers, 5=thefts/insecurity, 6=others (specify……………………………..)***
5. Have you and your household also lived in any other place? .... Yes / No
   (If 'yes': Go to 6; if 'no' go to Section C)
6. Where was that? [..]
7. What were your main economic activities in that place? [......................]
   (***1= farming, 2=fishing, 3=employment, 4=selling labour/ganyu, 5=others (specify…………..)***
8. When did you move to this place? [.......................] ***year***
9. When did you leave that place? [.......................] ***year***
10. Why did you leave that place? [........] (***As in 4***)

C) HOUSEHOLD CHARACTERISTICS

1. Household size (*number*) Male [ ] Female [ ]

2. Household information: Ethnicity […………………] Religion [……………………]

Name of HH Member (Start with the name of HH head)	Relation to HH head[1]	Sex[2]	Marital Status[3]	Age	Education[4]	Can read /write[5]	Occupation[6]	
							Primary	Second
1.								
2.								
3.								
4.								
5.								
6.								
7.								
8.								
9.								
10.								
11.								
12. Number of other regular members =								

[1] Relation to HH Head:	***01= H.H. Self, 02=Wife, 03=Husband, 04= Son, 05=Daughter, 06= Father, 07=Mother, 08=Daughter in law/son in law, 09=Brother, 10=Sister, 11=Father in Law, 12=Mother in Law, 13= Nephew, 14= Grandfather 15=Grandmother, 16=Others (specify)………………….***
[2]Sex:	***1=Male, 2=Female***
[3]Marital Status:	***1=Unmarried, 2=Married, 3=Widow, 4=Divorced, 5=Separated***
[4]Education:	***1=Junior primary (std 1-5), 2=Senior primary (std 6-8), 3=Form 1-2, 4=Form 3-4, 5=University and any professional training, 6=Non formal education, 7=No education***
[5]Can read & write:	***1=Yes, 2=No,***
[6]Occupation:	***01=Unemployed, 02=Agriculture, 03=fishing, 04=fishing and Agriculture, 05= House helper/maid, 06=Professional (Blacksmith, Carpenter, Sewing, etc.), 07=Business (Specify..), 08=Housewife, 09= IGA at home, 10=NGO worker, 11= Private/govt. service, 12= Student,13= Others (specify).***

For all not applicable indicate - 99

3. Are there any absent household members? .... Yes / No
   (Determine whether or not to consider them part of the HH, using question 4, 5 and 6)

4. Why are they absent? [………] ***(1=seasonal labour migration, 2=education, 3=staying with family elsewhere, 4= start own household, 5 =others –specify***

5. Are they absent for a period longer than 6 months? .... Yes / No

6. If 'yes', are they part of a household in the place where they stay?...Yes / No
   (If 'yes': Do not consider as HH member)

7. Do some present HH members stay in the house for less than 6 months a year?...................... Yes / No
(Determine whether or not to consider them part of the household, using Question 8 and 9)

8. Why do they leave the house [……….]
(1=seasonal labour migration, 2=education, 3=staying with family elsewhere, 4= split-up household, 05= others -specify)?

9. Are they part of a household in the place where they usually go? .... Yes / No
(If 'yes': Do not consider as HH member) - (Make the decision about who to consider as part of the household)

10. How many people are part of this 'household'?...
(This will be the research unit for the rest of this questionnaire)

D) FARM CHARACTERISTICS AND LAND TENURE

1. Do you own land?... Yes / No

2. Do you farm?..Yes / No

3. Do you also farm land that you do not own? .... Yes / No (***If 'no': Go to 5)***

4. Under what arrangement do you use this land? [.......................................]
1=cash rent, 2=share harvest, 3=use at no cost, 4=others-specify

5. Do you farm all the land you own? .... Yes / No (***If 'yes': Go to 7)***

6. What do you do with the land you own and do not farm? [.................................]
1= rents it to others, 2=use it for animal grazing, 3= as fallow 4= stays idle, 5=others specify

7. Could you tell us the long-term changes over time of the following items:
(Compares with the time that household started - in household history)

Crops	Past (years)							Now					
	Area			Yield				Area			Yield		
Maize	+	±	–	+	±	–		+	±	–	+	±	–
Rice	+	±	–	+	±	–		+	±	–	+	±	–
Millet	+	±	–	+	±	–		+	±	–	+	±	–
Sorghum	+	±	–	+	±	–		+	±	–	+	±	–
Groundnuts	+	±	–	+	±	–		+	±	–	+	±	–
Cassava	+	±	–	+	±	–		+	±	–	+	±	–
Potatoes	+	±	–	+	±	–		+	±	–	+	±	–
Beans	+	±	–	+	±	–		+	±	–	+	±	–
Pigeon peas	+	±	–	+	±	–		+	±	–	+	±	–
Tobacco	+	±	–	+	±	–		+	±	–	+	±	–
Chick peas (Tchana)	+	±	–	+	±	–		+	±	–	+	±	–
Sunflower	+	±	–	+	±	–		+	±	–	+	±	–
Cotton	+	±	–	+	±	–		+	±	–	+	±	–
Other 1……………	+	±	–	+	±	–		+	±	–	+	±	–
Other 2……………	+	±	–	+	±	–		+	±	–	+	±	–
	+	±	–	+	±	–		+	±	–	+	±	–

+ = has improved, ± has remained the same, – has decreased

8. Farm Plots

Plot No.	Size (acres/ha - specify)	Land Form (a)	Use (past year) (b)	Soil type (c)	Fertility (d)	How acquired? (e)	Crops in order of importance	Planting (Month)	Harvesting (Month)	Amount harvested (kgs/bags)
1							1.			
							2.			
							3.			
							4.			
							5.			
2							1.			
							2.			
							3.			
							4.			
							5.			
3							1.			
							2.			
							3.			
							4.			
							5.			
4							1.			
							2.			
							3.			
							4.			
							5.			
5							1.			
							2.			
							3.			
							4.			
							5.			

(a) Land Form
1) Dambo/Dimba
2) Wetland
3) Upland

(b) Use (past year)
1) Food crop
2) Cash crop
3) Fallow
4) Not used
5) Given out
6) Other (specify)

(c) Soil type
1) Gravel
2) Sand
3) Loam
4) Clay

(d) Fertility
1) good
2) not good
3) medium

(e) How acquired?
1) Inherited
2) Purchased
4) Gift from VH
4) Rented
5) Others (specify)

9. Do you practice irrigation farming?..Yes/No

10. If Yes

Crop	Area	Purpose (a)

(a) Purpose: 1=sale, 2=food, 3= sale+food

E) LIVESTOCK

1. Do you own animals? ....Yes / No

2. Did you own animals in the past? ....Yes / No
   If 1 'yes' and 2 'yes' go to 4. If 1' no' and 2 'yes' go to 3. If 1 'no' and 2 'no' go to section F)

3. How did you lose your animals? [...................]
   1=diseases, 2=theft, 3=sold them all after crisis, 4=grabbed by debtors, 5=others (specify)

4. Are you sometimes forced to sell animals in order to support your family? ....Yes/No

5. If Yes what are the critical months this happen? ....

6. Why? [.................................]
   1=buy grain to feed the family, 2=get money to pay school fees, 3= get money to pay for initiation, 4=slaughter to hire farm labour, 5=others (specify

7. Animal Form

Type of livestock	**No. of animals**		**Use (a)**
	10 years ago	**Now**	
Chickens			
Guinea fowls			
Ducks			
Rabbits			
Goats			
Sheep			
Pigs			
Cattle			
Other 1……………………………			
Other 2……………………………			

(a). Use of livestock
1) consumption- meat/milk/eggs
2) Source of income –meat/milk/eggs
3) Animal traction
4) Manure
5) Social obligations
6) other (specify)

F) FOOD SECURITY

How many months the household feeds itself adequately either from their own production, or purchasing from the market, or from other sources?

	Months with adequate food (tick)											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1. From own production												
2. Buys from markets												
3. Months most difficult to find food												
4. Months you normally receive food aid												

5. How many years were you able to sell surplus grains in the past 10yrs (***specify period if new household?)*** ……………………..

6. Food consumption

Food	Peak	Lean
1. Carbohydrates		
a) Maize		
b) Rice		
c) Sorghum/millet		
d) Cassava		
2. *Proteins*		
e) Fish		
f) Chicken		
g) Wild birds		
h) Meat (goats/rabbits/beef)		
i) Eggs		
3. *Vitamins*		
j) Leafy vegetables		
4. Legumes		
k) Pigeon peas		
l) Beans		
m) Soya		
n) Chick peas (Tchana)		
5. *Oils*		
o) Cooking oil		
p) G/nut seasoning		
q) Sunflower seasoning		
6. Fruits		

Code:
1=3meals/day,
2=2meals/day
3=1 meal/day
4= 1-3 days/week,
5 = 4-6 days/week,
6=1-3 days/month
7=Never
8=Irregular

7. Did you get any food out of hunting/fishing .... Yes / No

8. If Yes, what are these foods……………………………………………………………………………..

9. Did you get any food out of gathering *(specify)*?... Yes / No

10. Do you work on other people's farms in exchange for food or any other goods? .... Yes / No

11. If yes, When [………………..], Where [……………………], Season […………………………]

G) ACCESS TO NATURAL RESOURCES

1. What are the natural resources you have accessed in last 12 months

Resource type	**When available**												**How accessed (a)**
	J	**F**	**M**	**A**	**M**	**J**	**J**	**A**	**S**	**O**	**N**	**D**	
Forest mushrooms													
Firewood													
Poles													
Wild fruits													
Wild birds													
Fish & other aquatic products													
Marshes													
Others………………………													
Others………………………													

a) 1=open/common access, 2=buys license, 3=free permission from managers, 4= from own resource 5=tips govt managers, 6=other

2. How has the quantity accessed of these natural resources changed of the past 10 years(***specify period if new household?)***?

Resource type	**Amount accessed over time**		
Forest mushrooms	+	±	–
Firewood	+	±	–
Poles	+	±	–
Wild fruits	+	±	–
Wild birds	+	±	–
Fish & other aquatic products	+	±	–
Area under Marshes	+	±	–
Others………………………	+	±	–
Others………………………	+	±	–

+ = increased, ± = remained the same, – = decreased

3. Are there traditional beliefs connected with accessing natural resources? …………………..Yes/No
If Yes

Belief	**Natural Resource**
1.	
2.	
3.	

4. What are your sources of fuel?

   a) 1= Forest reserves, 2=Community woodlots, 3=own land/private source, 4=others specify
   b) Try investigating the best way respondents provides the amounts – either cost or cubic meters, please specify

Resource	**Source (a)**	**Amount used/ week (b)**
Forest wood		
Charcoal		
Planted trees		
Trees of Pigeon peas		
Trees of sunflower		
Maize stalks		
Rice Shaff		
Electricity		
Paraffin		
Others		

H) INCOME GENERATING ACTIVITIES

1. What are the sources of your household gross income? (tick)

   Once you have identified the income sources, ask the HH to rank 4 top contributing sources and write them in the extreme right column.

1	Farming		1st	
2	Livestock rearing		2nd	
3	Poultry rearing		3rd	
4	Fishing		4th	
5	Fish culture			
6	Bird hunting			
7	Agriculture wage labour			
8	Non agri. wage labour			
9	Petty business			
10	Business			
11	Urban remittance			
12	House helper/maid			
13	Handicraft			
14	Others (specify)			
15				

2. Did you get any goods (incl. foodstuff) by exchanging them for other goods (bartering) in the past 12 months?.. Yes / No

3. *If yes:* Which goods did you give and which goods did you get?
   Given ....
   Received:………………………………………………………………………………...

4. Has the number of income sources for your household increased, decreased or stayed the same over time (describe the period)?
   Increased [..........] Decreased [.....................] the same [..............]

5. Has your nonfarm income increased, decreased or stayed the same over time? (tick)
   Increased [..........] Decreased [.....................] the same [..............]

6. In which sector do women work outside home? *(1= always, 2=Sometimes, 3=Only in crisis, 4=never)*

Agriculture labour	
Fish processing and/or trading	
Firewood selling	
Food selling	
Craft making	
Fetching water for money	
Employment	

Selling labour (ganyu)	
Others Specify)	

I) CASH EXPENDITURE

1. What are the major areas of expenditure over the last 12 months *(use the form as checklist – use a convenient recall period – wk, month, year and specify)*

Expenditure details		**Estimated Cost**
Major foods	Maize/Rice/Cassava	
	Meat/eggs	
	Fish	
	Vegetables	
Other foods	Snacks	
	Tea (tea rooms)	
Leisure/beer		
Education		
Health		
Consumables	Clothes	
	Paraffin	
	Salt/sugar/…	
Firewood		
Transport		
Weddings/funerals/initiations		
Gifts		
Housing: repairs/improvements		
Investments	Agricultural inputs	
	Business	
	Equipments	
Loan repayments		
Others:		

2. Which types of expenditure have increased most sharply over time?
....
....

J) SAVINGS

1. Did you or do you have savings?.. Yes/No
(If no go to section K)

2. Have you used the money in last 2 years that you have saved?..Yes/No
(If no go to section K)

3. If Yes what have you used your savings for:

1.
2.
3.
4.
5.

K) DETAILS OF LOAN

1. Do you have any loans or have you borrowed money in last 12 months?...................................Yes/No
(If no go to section L).

2. If yes:

Sources	Money lender	
	Commercial banks	
	Community revolving funds	
	Microfinance Institutions (Specify)	
	Friends/relatives	
	NGOs	
	Clubs/CBOs	
	Others (Specify)	

How have you used the loan?	Farming	
	Off farm IGAs	
	Health	
	Housing	
	Emergency	
	Consumption	
	Others (Specify)	

3. What was the repayment period of the loans? and what was (the annual) interest rate? *(from 2)*

Loan	Repayment period *(specify wks/months)*	Interest

L) POSSESSIONS/ASSETS

1. Indicate whether the household possesses the following items, how many and how acquired?

Asset	No.	Acquired	Asset	No.	Acquired
Car/Motorcycle			Plough		
Bicycle			Fishing boats		
House –iron roofing			Seine nets		
TV set			Livestock		
Radio			Modern furniture		
Sewing machine			Treadle pump		
Ox cart			Others:		
Cell phone					

Acquired: 1= Inherited, 2= purchased by owner, 3=remitted by relations, 4= given by well-wishers, 5=Others,

2. If the household possesses a radio,

What are the favourite channels/radio stations	Programs	What times of the day do you normally listen to the radio

3. Are you sometimes forced to sell possessions because you need the cash?.............................. Yes / No
   (If no: Go to 5)

4. Have there been years that you were forced to sell much more possessions than usual…........ Yes / No

5. Have your possessions increased, decreased or stayed the same over time? (***indicate years***)
   Increased [................] decreased [...................] the same [……………]

M) MIGRATION

1. Have any members of this household left the area for over a month in the past year? .... Yes / No
   If 'yes': Go to first form; If 'no': go to Question 2

Name of migrant	Destination	HH status on food/cash /labour when leaving	Reason for leaving	Period (months)

2. Have any members of this household left the area for over a month in the **years**? .... Yes / No

If 'yes': Go to second form; If 'no': go to Section M)

Name of migrant	Destination(s)	HH food/cash /labour situation when leaving	Reason for migrating	How often in past 10 years

3. What were the main benefits the migrants brought to the household? [ ]
   (1=cash remittances, 2=food provision, 3=other)

4. Has the importance of migration and remittances for the household increased, decreased or stayed the same over time? Increased [.............] Decreased [..........] the same [............]

N) SOCIAL NETWORKS

		In the village			Outside Village			Outside Malawi		
1. Do you have relatives *(1=yes, 2=No)*										
2. Do you help each other in **farm** and/or other **work**? *(1=yes, 2=No)*										
3. Do you give or receive **food** to/from these relatives? *1=give, 2=receive, 3=No (neither gives nor receives)*										
4. Do you give or receive **cash** to/from these relatives *1=give, 2=receive, 3=No (neither gives nor receives)*										
5. Have these forms of mutual aid increased, decreased or stayed the same over time?	farming	+	±	–	+	±	–	+	±	–
	Food	+	±	–	+	±	–	+	±	–
	cash	+	±	–	+	±	–	+	±	–

6. Household membership in local institutions

Household member/s *(indicate number)*	**Affiliation Type**	**Affiliation** (tick)	**Benefits (a)**
	Affiliation with political party		
	Membership in community committees (school, church, etc - specify committee/s)		
	Membership in governance committees (VNRMC, VDC, ADC, BVC) - specify		
	Membership of church/mosque		
	Membership in NGO supported groups (farmers club, irrigation, AIDS etc) specify		
	Membership in CBO groups (voluntary)		
	Participation in community festivals		
	Other associations (specify)		

(a) **Benefits***: 1=get help in times of problems, 2=satisfaction in helping others, 3=Recognition, 4=security, 5=social collateral, 6=requirement by donors, 7 others – specify*

7. If household belongs to a donor supported group, does it receive subsidized inputs, or goods from NGOs through the group……………………………………………………………………Yes / No

8. What other benefits do you get from the donor/NGO?
   ……………………………………………………………………………………………………
   ……………………………………………………………………………………………………
   …………………………………………………………………………………………………….

9. Do you know other households that do not belong to any donor supported group…………… Yes / No
   If Yes How many […………….]

10. What is their economic status as compared to you?
   Better [………..] same […………..] Worse […………..]

O) VULNERABILITIES & COPING STRATEGIES DURING PAST 12 MONTHS

1. What kind of crisis did you experience in the last 12 months? (*tick*)

Flood/Excess rain	
Drought/ Little rain	
Cyclone /Wind damage	
Poor production/ Shortage of food	
Illness	
Death of household member	
Arrest of household member	
Divorce/separation	
Loss of job/ loss of reliable income source	
Theft/burglary	
Eviction	
Property grabbing	
Market fluctuation	
Accident of HH members	
Others (Specify	

2. What were your coping strategies?

Cash loan from neighbors/relatives	
Cash loan from Money Lender	
Food assistance from kins	
Reduced meal frequency	
NGO relief	
Government relief	
Sold/rented land	
Sold household in-house assets	
Sold livestock	
Sold firewood from forest reserve	
Sold grain – rice, maize	
Sold family labor for cash or food	
Changed occupation	
Migrated to sale labor	
Collected leftover grain from paddy field	
Others (Specify	

3. Past health status of household members

Type	Number of days			
	Last 1 Month		Last 12 months	
	Male	Female	Male	Female
Cholera				
Other diarrhea episodes				
Respiratory illnesses				
Malaria				
Other chronic illnesses				
Number of work days missed due to illness				
Number of times HH visited a doctor				
Number of times HH consulted traditional doctors				
Number of times HH visited faith healers				
Amount spent in last month on medicine & fees (MK)				

Indicate a cumulative number of all household members– so if one member 2 days and another 7, answer is 9 days.

4. Do you have access to water for: drinking? ☐ Washing? ☐
(*1=Yes, 2=No*)

5. If yes, what are the sources for: drinking? ☐ Washing? ☐
Source: 1=protected wells, 2=piped water, 3=river, 4=Lake Chilwa, 5=boreholes

THANK YOU - YOU HAVE FINISHED
THANK THE RESPONDENTS

Appendix 4: Research Team

Researchers

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2. Regson Chaweza
3. Precious Mwanza

Qualitative Research Team

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2. Andrew Zulu
3. Simu Ndanga (Miss)
4. MacDonald Chitekwe

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Data Entry

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